

A trend analysis across five *Drosophila* EM connectomes
and five segmentation error models

BANC · FAFB · MANC · MCNS · MAOL

5 error models × 10 error rates × up to 5 replicate trials → 1,030 trials

Framework v1.0.0 · every number recomputed from the per-trial CSVs

11 August 2026

Contents

How this report is organised

This report analyses how five segmentation error models perturb the structure of five FlyWire connectomes as the error rate grows from 0 % to 20 %. Every figure and every number was recomputed from the per-trial result files.

- §1 Research question and one-sentence answer.
- §2 The data — five connectomes, five error models, coverage and rate grids.
- §3 Method — how each error model works and how effects are measured.
- §4 Verification — consistency checks before analysis.
- §5 Results — one figure per error model, PageRank, heatmap and effect tables.
- §6 Conclusions — practical implications for segmentation benchmarking.
- §7 Reproducibility — commands to regenerate every figure and table.
- Appendix — statistical notes on known artefacts and partial runs.

§1 · Research Question

What this report answers and what you will find

Automated segmentation of electron-microscopy data is imperfect. Reconstruction errors are classified into five types: missed synapses (false negatives), false synapses (hallucinated connections), imprecise synapse-count measurement, split neurons (over-segmentation) and merged neurons (under-segmentation). Each corrupts the wiring diagram differently: some remove connections, some add them, some change only connection strengths, and some change the very identity of neurons.

This report asks: as the error rate grows from 0 % to 20 %, how strongly does each error type perturb the global structure of five FlyWire connectomes — and are those effects consistent across datasets?

Executive Summary

Topology-deleting errors (missed synapses, merge mis-segmentation) exert the most severe structural impact, scaling almost perfectly with the error rate. Measurement noise remains remarkably benign. Furthermore, all evaluated datasets exhibit qualitatively consistent responses, indicating that these structural responses are consistent across the evaluated connectomes.

Core Investigative Dimensions

- Direct: how do edge count, synapse count and degree respond as the error rate rises?
- Structural: are connectivity, component sizes and hub rankings preserved, or do they degrade?
- Comparative: which error type is most disruptive at the same nominal rate — and which is benign?

§2 · The Data

Five connectomes, five error models, 1,030 trials

The analysis covers five *Drosophila* EM connectomes: BANC (brain + nerve cord), FAFB (full adult female brain), MANC (male adult nerve cord), MAOL (male adult optic lobe) and MCNS (multi-cell-type nervous system). Each dataset was perturbed with up to five error models at ten error rates, with five independent seeded trials per rate, and every trial ran six graph analyses.

Table 1 · Baseline network properties at 0 % error (identical across all error models and trials)

Dataset	Neurons	Edges	Total synapses	Density	Reciprocity	Assortativity	Largest WCC
BANC	158,262	3,990,039	23,556,214	0.00016	0.130	-0.046	153,952
FAFB	139,255	5,342,446	50,666,648	0.00028	0.165	-0.036	136,997
MANC	23,665	6,239,883	30,934,610	0.01114	0.285	+0.037	23,641
MCNS	166,694	6,239,112	89,797,169	0.00022	0.157	-0.057	165,820
MAOL	52,445	6,736,968	26,492,194	0.00245	0.278	-0.054	51,503

§2.2 · Data Coverage and Rate Grids

Which error models were run on which datasets

Table 2 · Replicate trials available per error model (×5 trials per rate; — = not run)

Dataset	Missed syn.	False syn.	Syn. count	Split	Merge
BANC	5x5	5x5	5x5	5x5	5x5
FAFB	5x5	5x5	5x5	5x5	2
MANC	5x5	—	5x5	5x5	1
MCNS	5x5	5x5	5x5	5x5	5x5
MAOL	5x5	—	5x5	5x5	—

Three error-model runs are missing or partial: MANC and MAOL have no false-synapse run, MAOL has no merge-error run, FAFB merge has 2 trials/rate and MANC merge has 1 trial/rate. All other cells have the full 5 trials.

Two rate grids are used: missed/false synapses at 0, 0.25, 0.5, 0.75, 1, 2, 5, 10, 15, 20 %; synapse-count, split and merge errors at 0, 0.5, 1, 2, 3, 5, 7.5, 10, 15, 20 %. All values are means over the replicate trials.

What the six analyses measure

- **basic_structure** — node count, edge count, total synapses, weight mean/variance/std/max, density.
- **degree_distribution** — in/out/total degree mean, std and max.
- **pagerank** — weighted PageRank (damping 0.85); Pearson/Spearman correlation and top-100 overlap of perturbed vs baseline scores.
- **assortativity** — degree assortativity of the directed graph.
- **connected_components** — weak and strong component counts and largest sizes.
- **reciprocity** — fraction of edges with a reciprocal counterpart.

§3 · Method

How each error model perturbs the connectome graph

Table 3 · The five error models and what they do

Error model	Perturbation applied	Expected structural effect
Missed synapses	Removes synapses per edge with calibrated probability; an edge is deleted only when its weight reaches 0.	Edges ↓, weights ↓, degree ↓
False synapses	Adds $k = \text{round}(\text{rate} \times \text{edges})$ new edges sampled from region-restricted, Jaccard-ranked candidate pairs; weights from the weak-edge distribution.	Edges ↑, density ↑
Synapse count measurement	Adds Gaussian noise to edge weights, $\sigma = \text{rate} \cdot w$, clamped ≥ 1 ; topology untouched.	Weights only; structure unchanged
Split errors	Splits $\text{round}(\text{rate} \times \text{eligible})$ neurons (total degree ≥ 10) into two fragments; every edge is rewired exactly once.	Neurons ↑, edges preserved, degree ↓
Merge errors	Merges $k = \text{round}(0.5 \cdot \text{rate} \cdot \text{eligible})$ region/soma-compatible neuron pairs ranked by shared partners; edges re-attached, parallels summed, $A \leftrightarrow B$ dropped.	Neurons ↓, edges collapse, synapses $\approx$ conserved

What the nominal rate r means: the models are not directly comparable at the same r , because r is a different biological quantity in each model. Missed synapses remove a fraction r of synapses (removal probability is vulnerability-weighted, so per-edge removal is not uniform, but aggregate synapse loss matches r exactly). False synapses add $k = \text{round}(r \times \text{edge count})$ new edges. Synapse-count measurement applies proportional noise $\sigma = r \cdot w$ to every weight. Split errors split a fraction r of eligible neurons (total degree ≥ 10). Merge errors absorb a fraction r of candidate-pool neurons into $k = \text{round}(0.5 \times r \times \text{eligible})$ pairs. All stochastic operations use recorded seeds, so every trial is exactly reproducible.

§3.2 · Effect Measure and Scale Conventions

How to read every trend figure in this report

How to read the figures

Every trend figure in §5 uses a unified layout for cross-dataset comparison: error rate (%) is plotted on the x-axis, and the relative percentage change of a metric against its 0 % baseline is on the y-axis. The formula is $\Delta\% = (m(\text{rate}) - m(0)) / |m(0)| \times 100\%$. A value of -20% indicates a 20 % reduction from the baseline value. The dashed horizontal line marks zero change (no effect).

Scale conventions

- All evaluated connectome datasets are overlaid on a single axis for direct comparison.
- Direct endpoint labeling replaces traditional legends; dataset names and terminal values are positioned adjacent to each line.
- Automatic displacement prevents label occlusion when final values converge.
- PageRank is plotted as Pearson correlation (not % change) because the baseline is exactly 1.0. Coloured bands show quality thresholds.
- For split/merge errors the node set changes; PageRank vectors are re-aligned to baseline node space before correlating.

§3.3 · Statistical Caveats

Reading the trend figures and handling artefacts

Figure conventions

Figures in §5 use error rate (%) on the x-axis and relative % change $\Delta\%$ against the 0 % baseline on the y-axis, calculated as $\Delta\% = (m(\text{rate}) - m(0)) / |m(0)| \times 100 \%$. All datasets are plotted on unified axes to facilitate direct comparative analysis.

Statistical caveats

- Baseline Inflation: Assortativity has a near-zero baseline, so % change is inflated. We report it but exclude it from the heatmap.
- Metric Scaling: metrics whose absolute change stays below 0.25 % across all rates are treated as negligible. This is a descriptive magnitude threshold, not statistical significance.
- PageRank: Plotted as Pearson correlation (r) rather than % change. Pearson measures vector correlation, not rank preservation; Spearman correlation and top-100 overlap are reported in §5.5b.
- Split/Merge: PageRank vectors are re-aligned to baseline node space before correlating.
- Partial Runs: Cross-dataset means for merge errors include FAFB (2 trials) and MANC (1 trial).
- Uncertainty: trial-to-trial SD of the reported % changes is below 0.15 pp for every 5-trial cell; the MANC merge row ($n = 1$) has no trial variance.

§4 · Verification

Consistency checks performed before analysis

Table 4 · Data-quality verification

Check	What was tested	Outcome
Completeness	6/6 analyses present and SUCCESS in every trial	Pass — 6,180 analysis rows, 0 failures
Baseline invariance	0% runs identical across trials and error models	Pass — max rel. diff 0.0 (bit-identical)
Degree identity	$\text{sum}(\text{in-degrees}) = \text{edge count}$; $\text{in-degree mean} = \text{edges/nodes}$	Pass — max rel. err $2.2\text{e-}16$
Density identity	$\text{density} = \text{edges} / (n \cdot (n-1))$	Pass — max rel. err $7.0\text{e-}13$
Missed synapses	$\text{synapses removed} \approx \text{rate} \cdot \text{synapses}$ (QC enforced)	Pass — removal matches target
False synapses	edge additions match planned $k = \text{round}(\text{rate} \cdot \text{edges})$	Pass — at every rate, all trials
Syn. count noise	$\text{weight-variance increase} \approx \text{rate}^2 \cdot E[w^2]$ (theory)	Pass — predicted 4.98 vs observed ≈ 5.1 at 20%
Split/merge	edge count preserved (split) / collapse bookkeeping (merge)	Pass — matches plan metadata
PageRank alignment	split/merge vectors re-aligned to baseline node space	Pass — correlations in $[0.5, 1]$ at all rates

The two structural identities ($\text{in-degree mean} = \text{edges/nodes}$ and $\text{density} = \text{edges}/(n \cdot (n-1))$) hold to machine precision, and the 0 % baseline is bit-identical across trials and error models — so all observed changes below are caused by the perturbation, never by run-to-run variability.

§4.2 · Anomalies Investigated and Resolved

Suspicious patterns that were examined and explained

- 'False-synapse degree shift never varies across trials' — a mathematical identity, not a bug: false synapses add a deterministic number of edges every trial, so their trial-to-trial spread is exactly zero. Missed synapses remove a binomial-random number, so their spread is non-zero.
- 'Synapse-count measurement changes almost nothing' — by design it perturbs weights only; the variance increase matches theory ($\text{rate}^2 \cdot E[w^2]$), and structural metrics are completely untouched.
- 'Largest weak component is flat for false synapses' — new edges land inside the already-giant component, so adding them never enlarges it; the strong component does grow.
- 'Edge count is flat for split errors' — splitting rewires edges (each assigned exactly once) rather than creating them; the neuron count grows instead.
- No formula errors were found in the outputs. The only real caveat is run coverage: FAFB and MANC merge-error results are partial (2 and 1 trials per rate), and MANC/MAOL have no false-synapse run.

Conclusion: The datasets exhibit internal consistency and are structurally robust.

§5 · Results

Principal observations across all error models

Figures in this section show, for each combination of error model and metric, how that metric changes as a function of error rate. Each figure occupies one full landscape page for maximum readability: datasets are overlaid on a single axis with direct endpoint labeling to enable immediate cross-dataset comparisons. Lines approaching the zero axis indicate high structural resilience.

Principal observations (see Table 5)

- Missed synapses: synapses fall one-for-one with the rate (−20 % at 20 %); edges only −5 %; PageRank Pearson r 0.999.
- False synapses: edges and mean degree rise in step with the rate (+19 % at 20 %); PageRank Pearson r 0.994.
- Synapse-count noise: only the weight variance moves (+5.5 %); all topological metrics are untouched.
- Split errors: edges preserved exactly, but spread over more neurons → mean degree −15 %, largest weak component +18 %.
- Merge errors: most invasive to connectivity — edges −11 %, weight variance +47 %, largest weak component −9 %.

Missed Synapses — Edge count

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

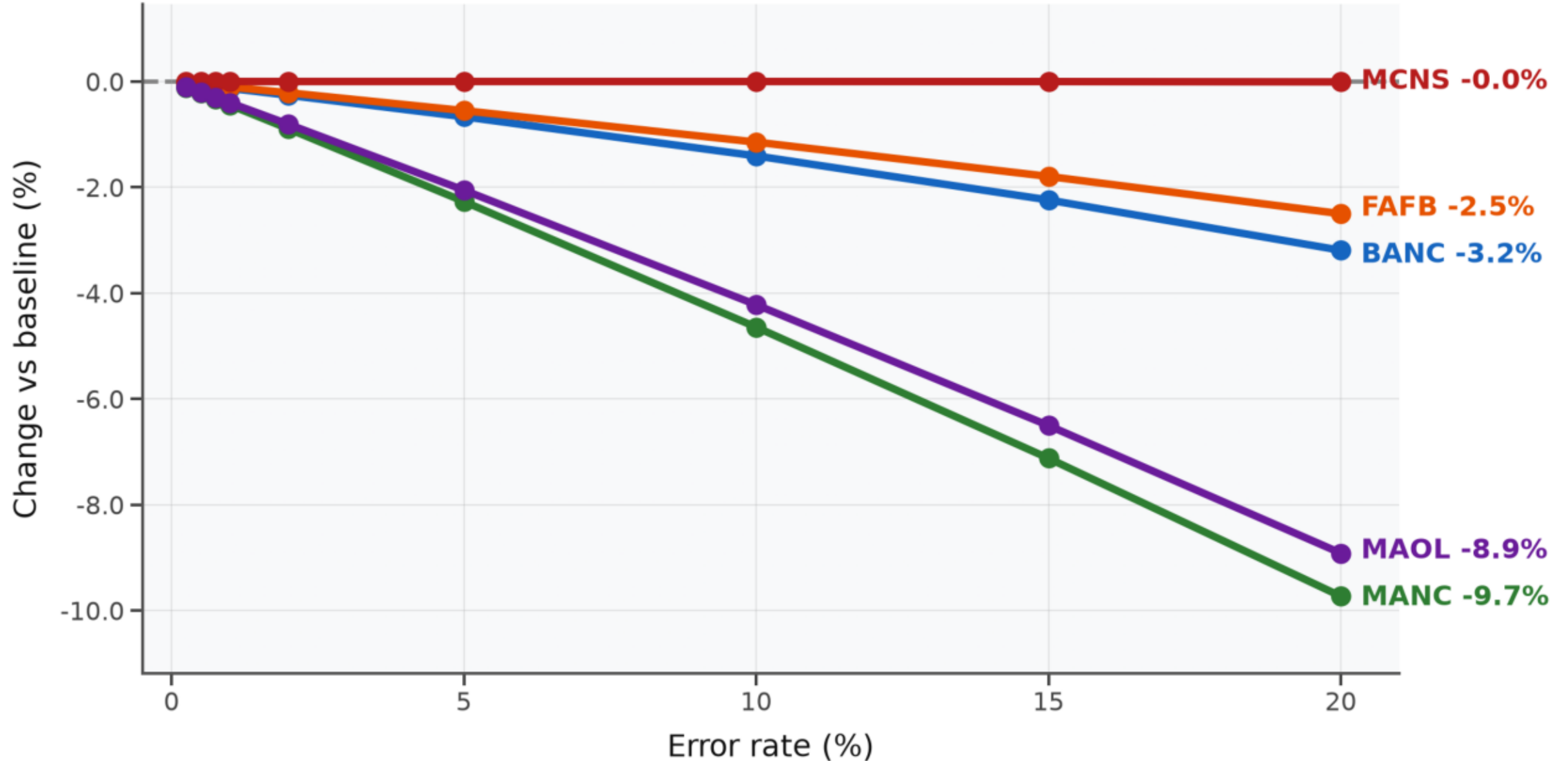


Figure 1 · Missed synapses — Edge count. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Missed Synapses — Total synapses

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

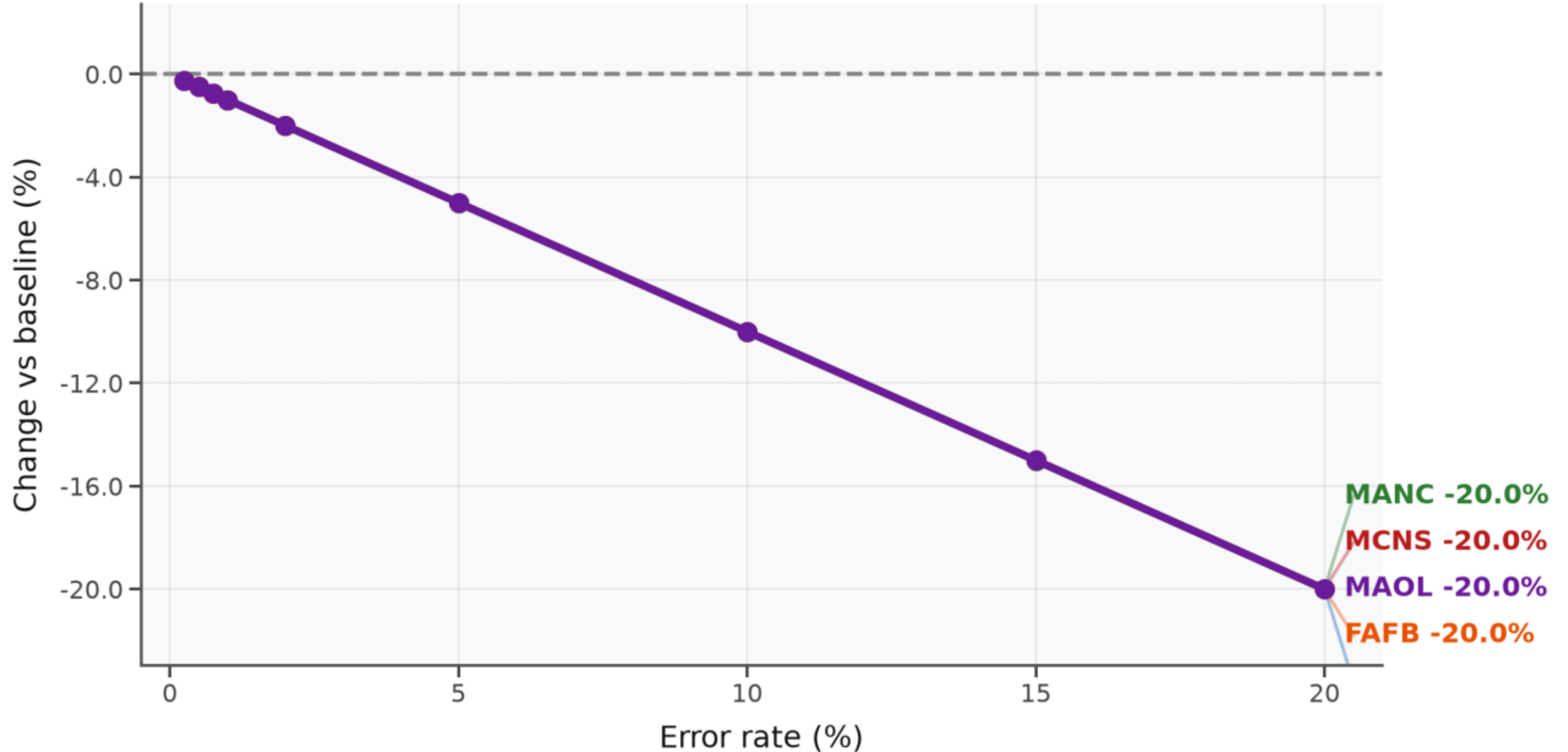


Figure 2 · Missed synapses — Total synapses. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Missed Synapses — Mean total degree

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

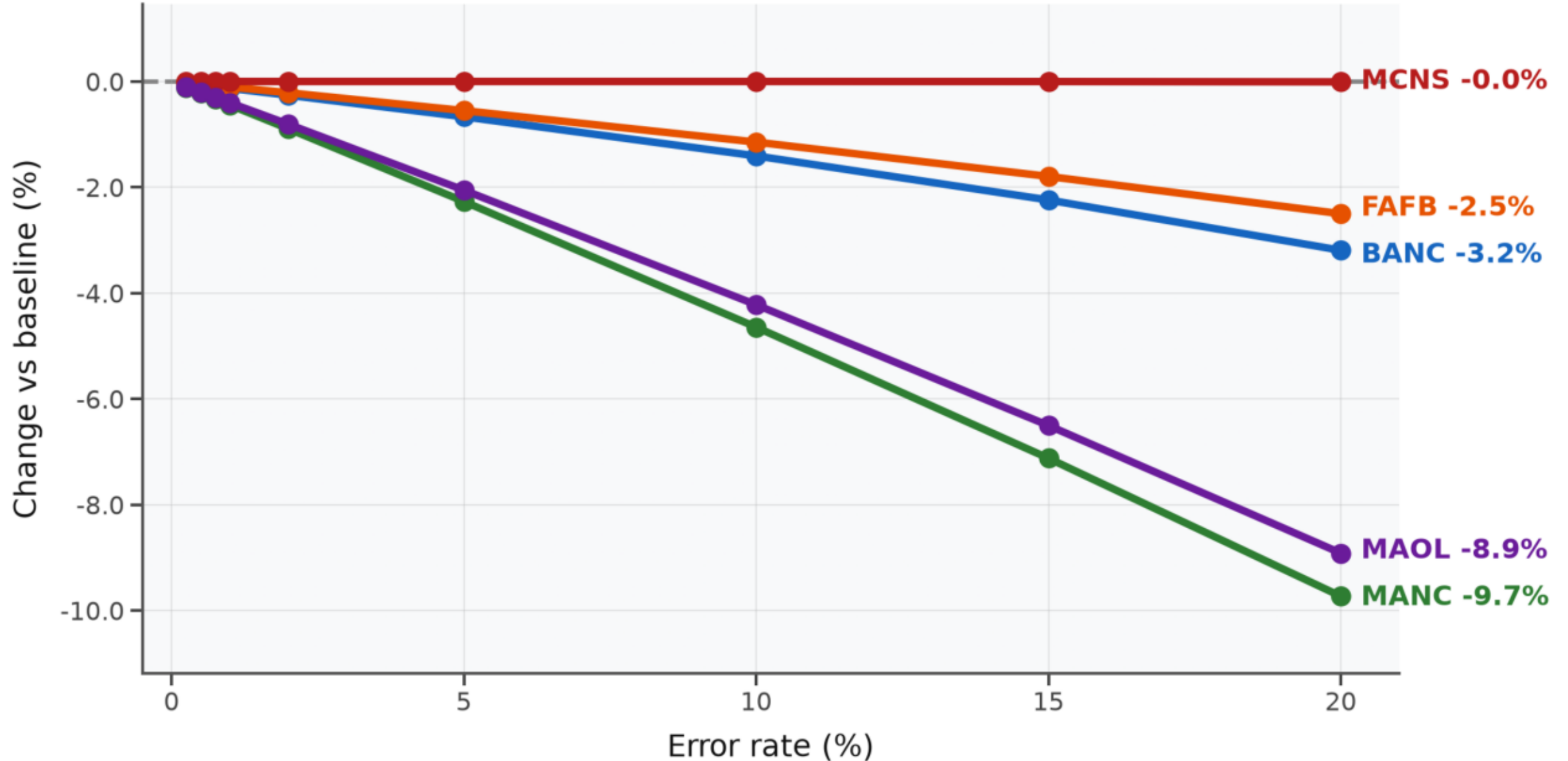


Figure 3 : Missed synapses — Mean total degree. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Missed Synapses — Largest weak component

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

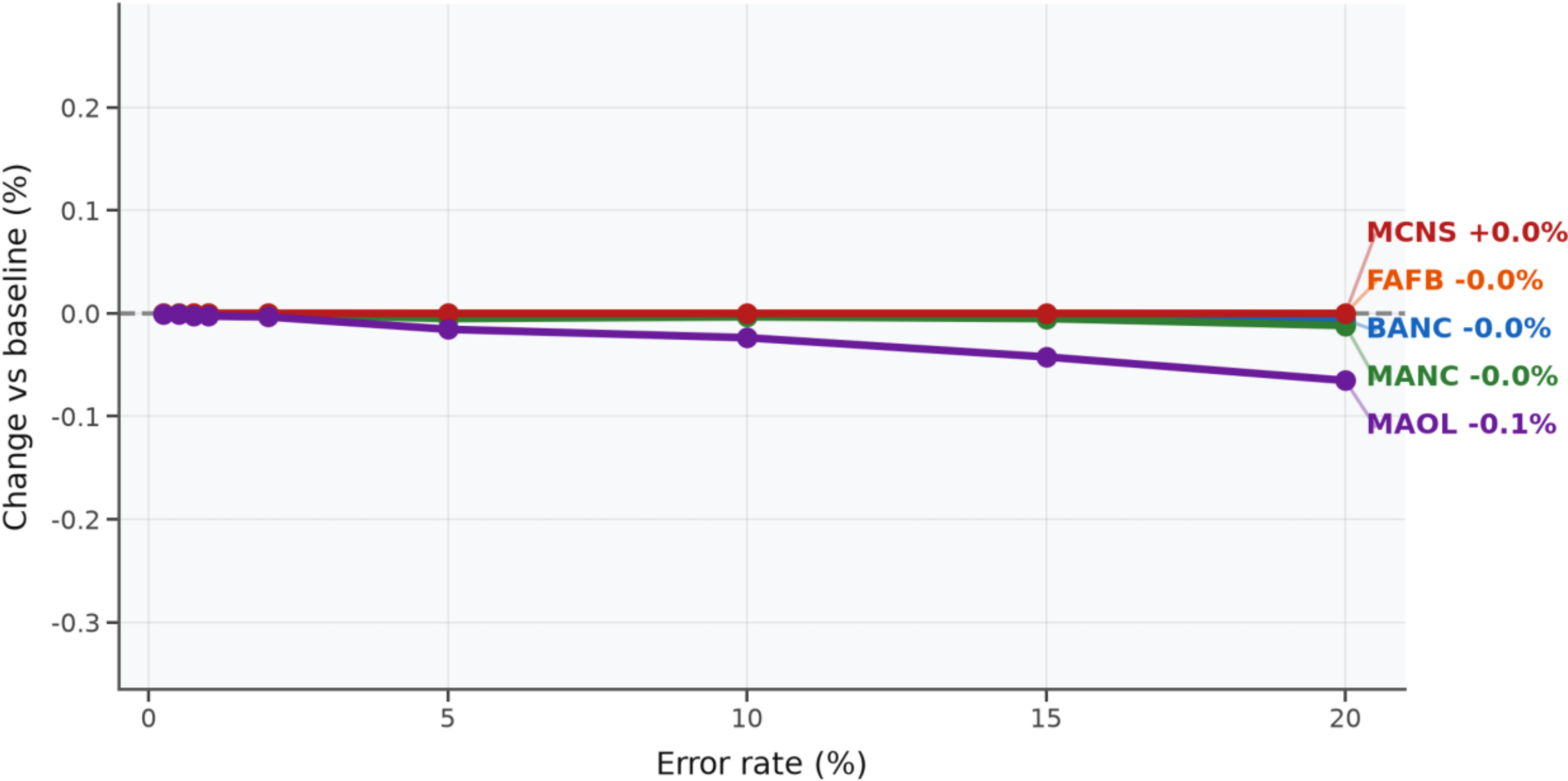


Figure 4 · Missed synapses — Largest weak component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Missed Synapses — Reciprocity

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

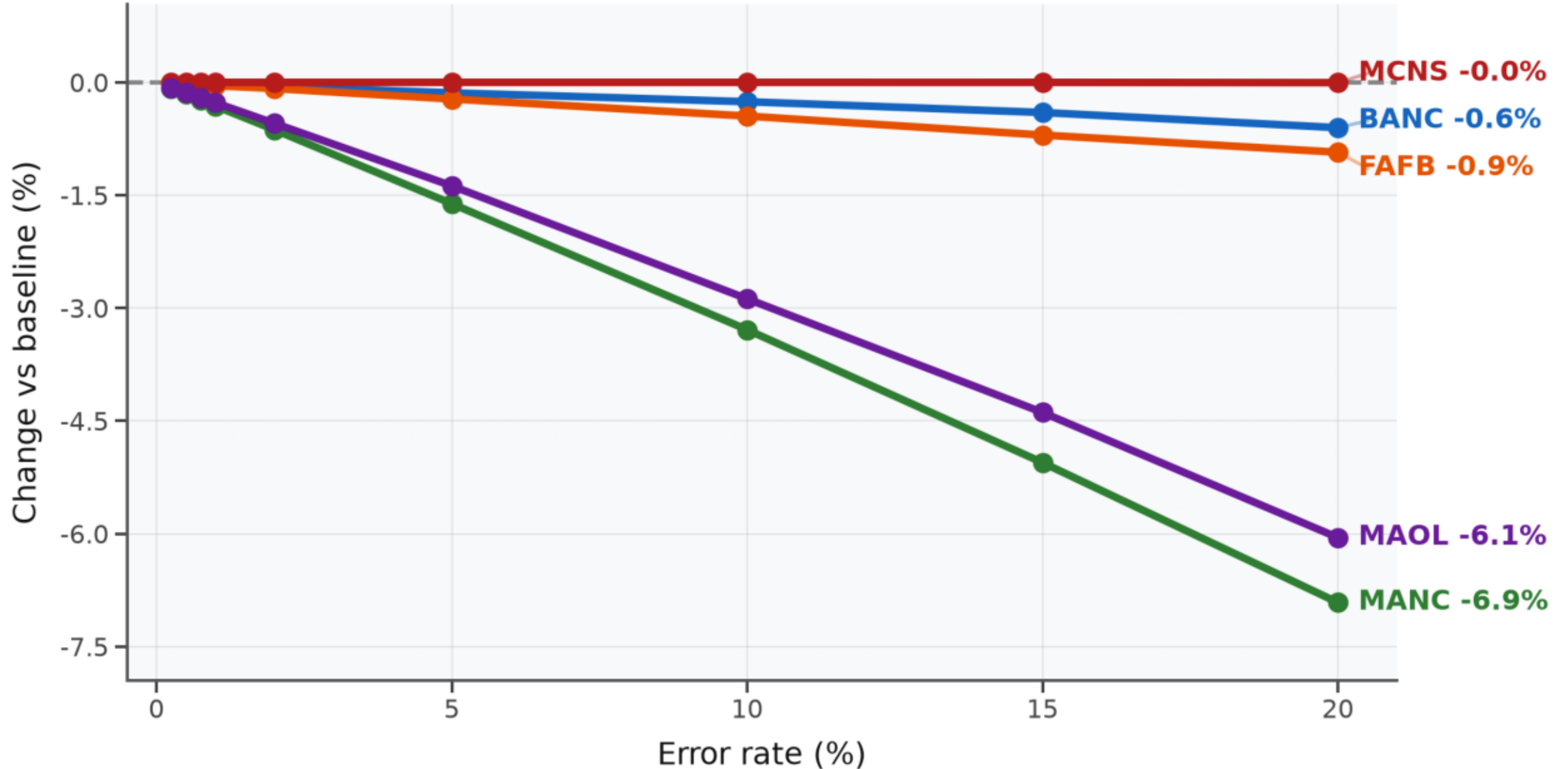


Figure 5 · Missed synapses — Reciprocity. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

False Synapses — Edge count

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

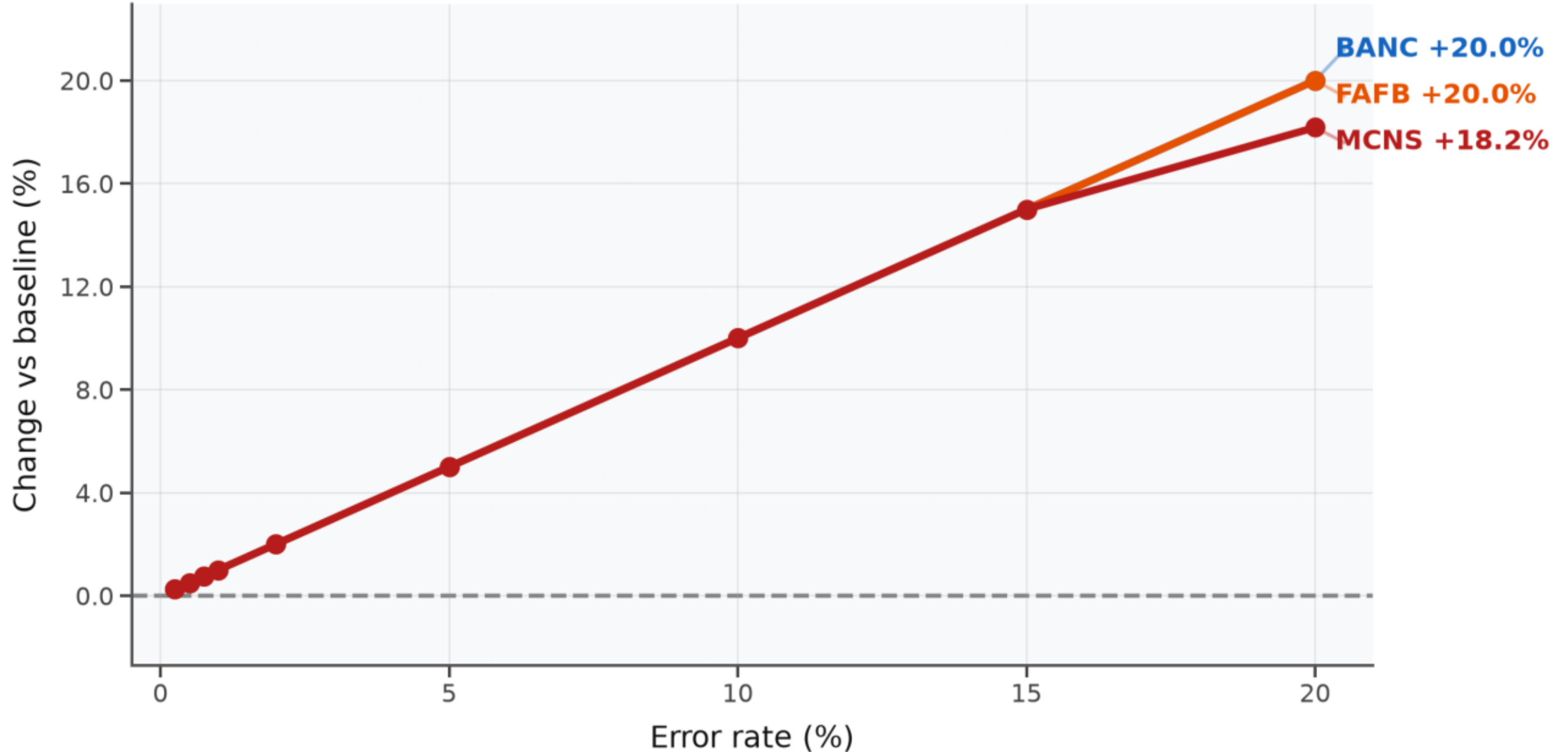


Figure 6 · False synapses — Edge count. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

False Synapses — Total synapses

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

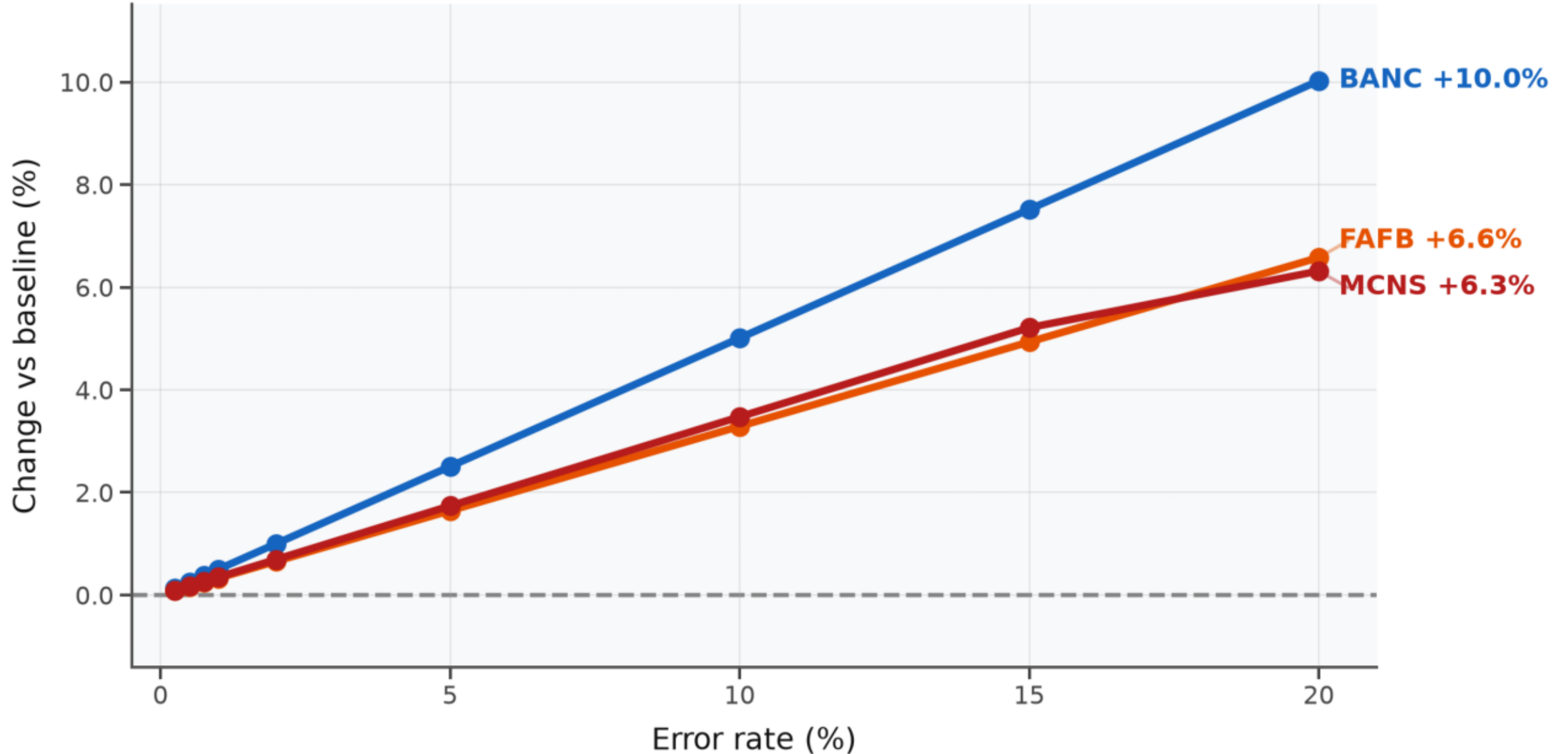


Figure 7 · False synapses — Total synapses. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

False Synapses — Mean total degree

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

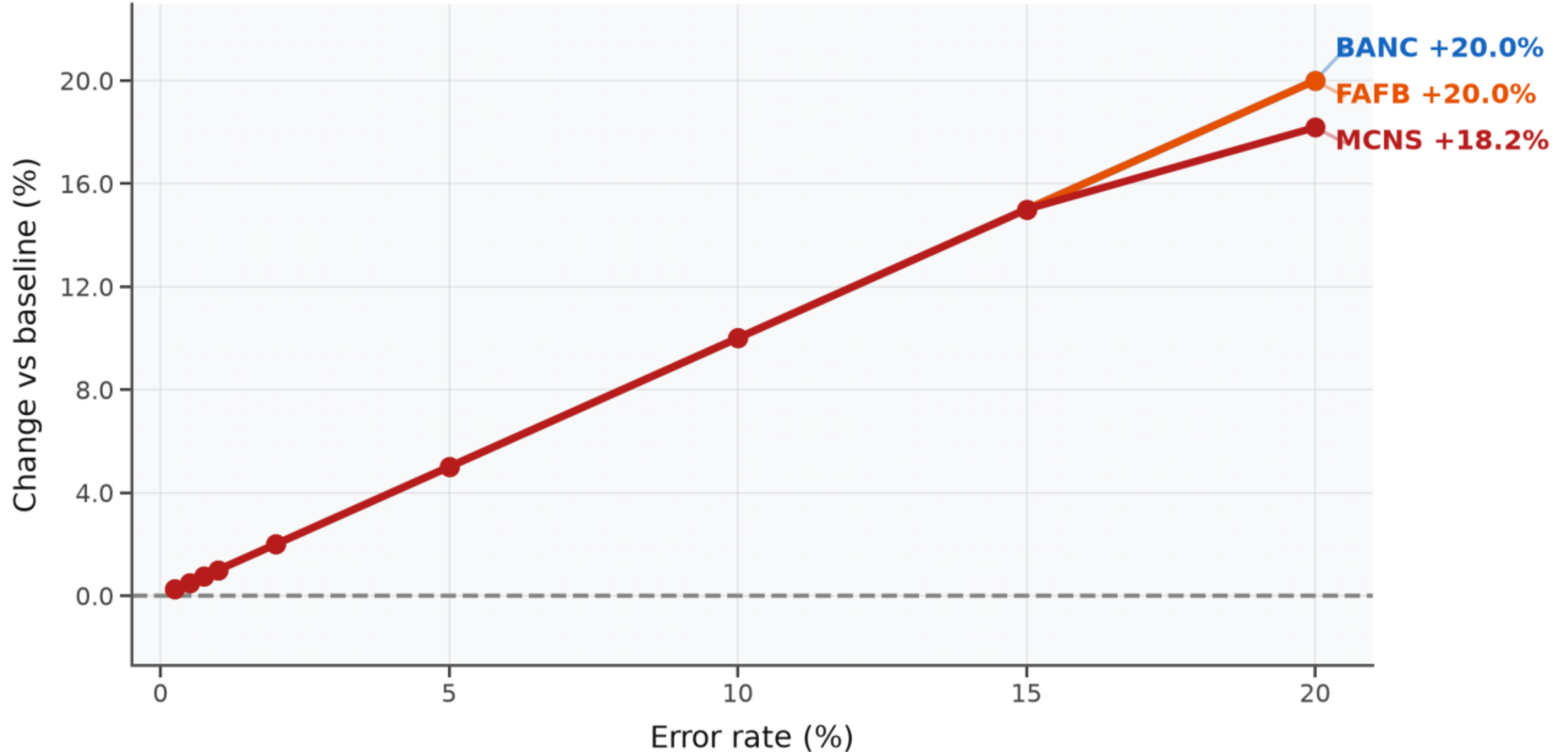


Figure 8 : False synapses — Mean total degree. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

False Synapses — Largest weak component

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

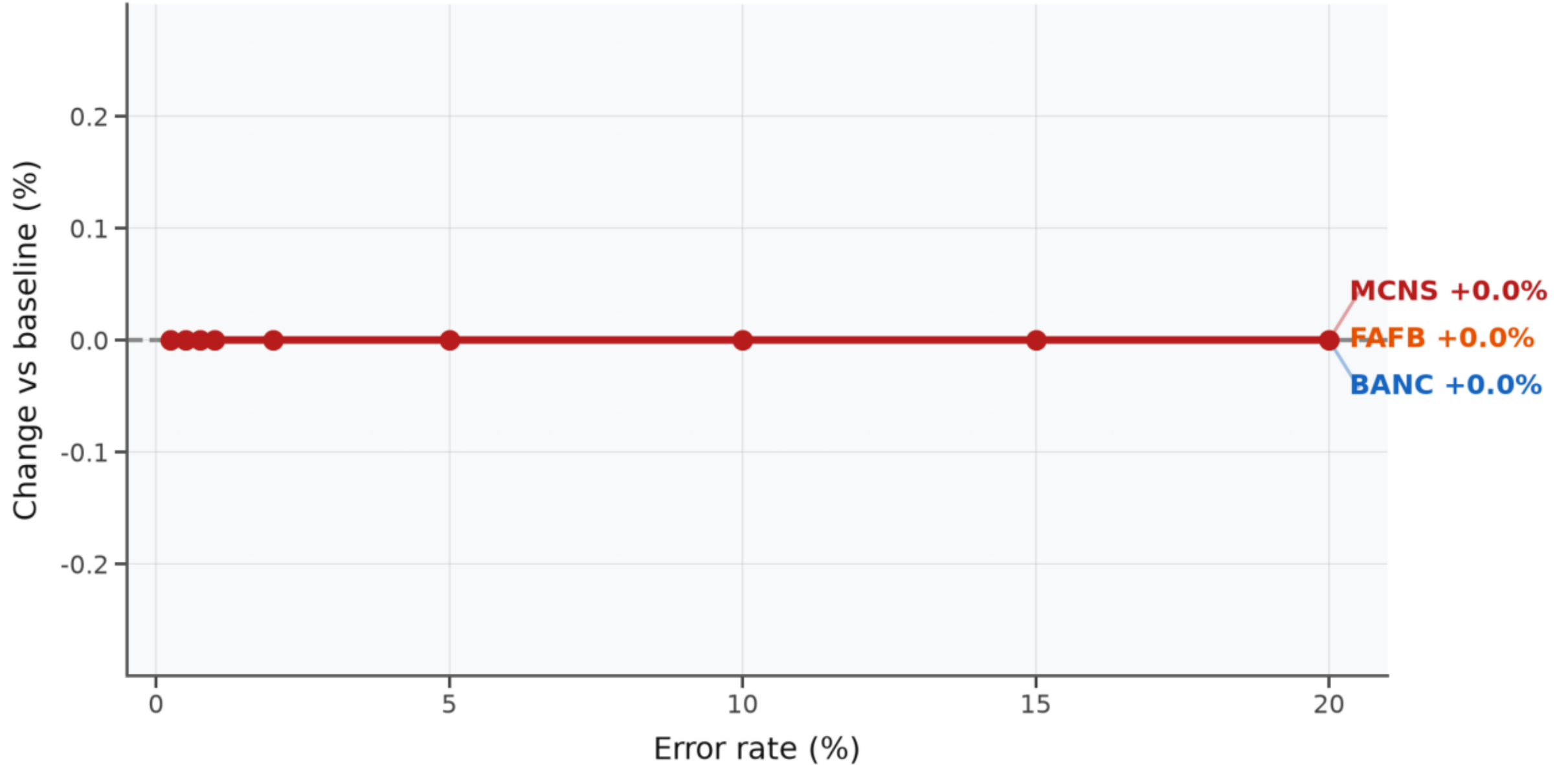


Figure 9 · False synapses — Largest weak component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

False Synapses — Reciprocity

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

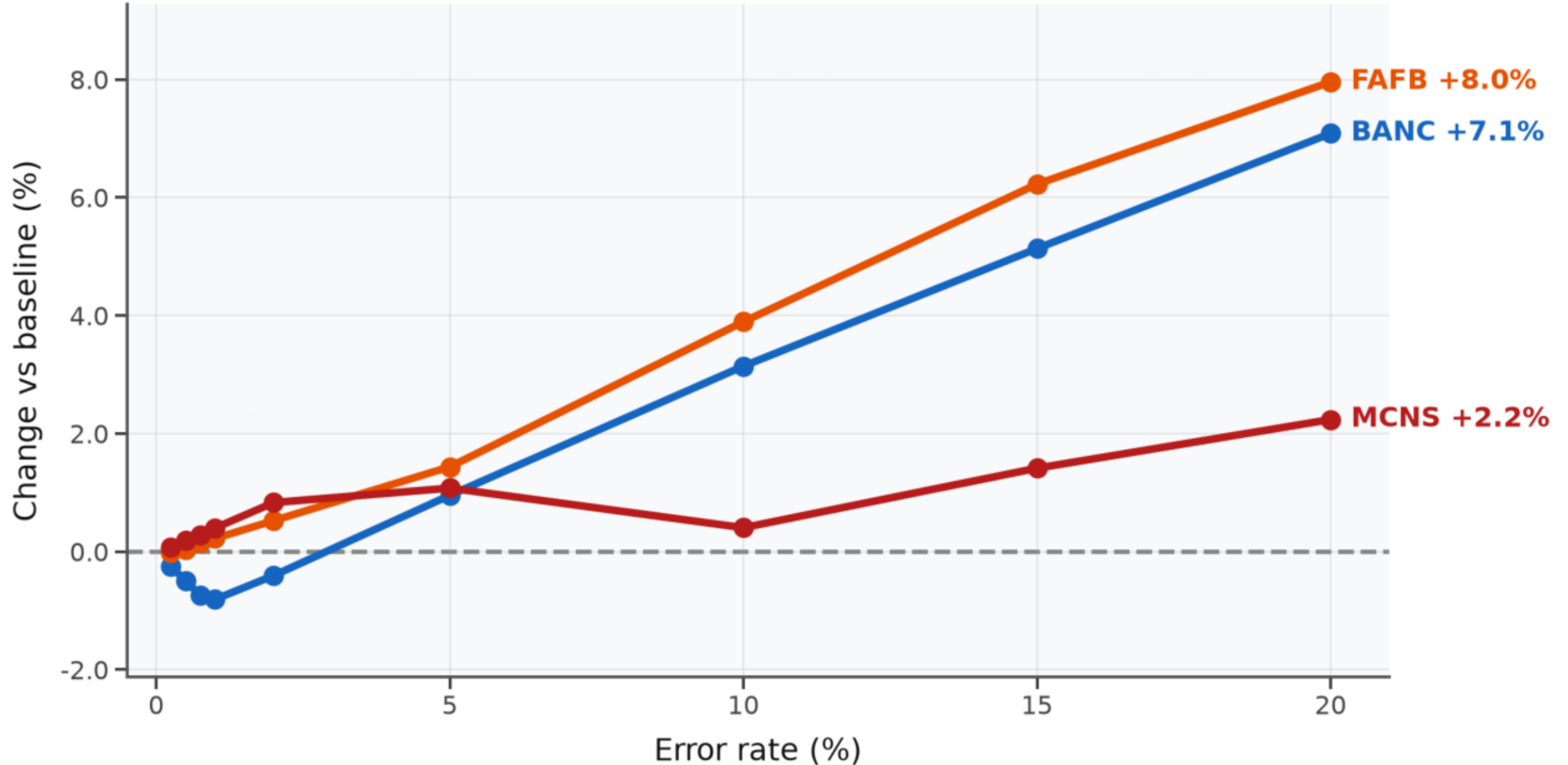


Figure 10 · False synapses — Reciprocity. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Synapse Count Noise — Edge count

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

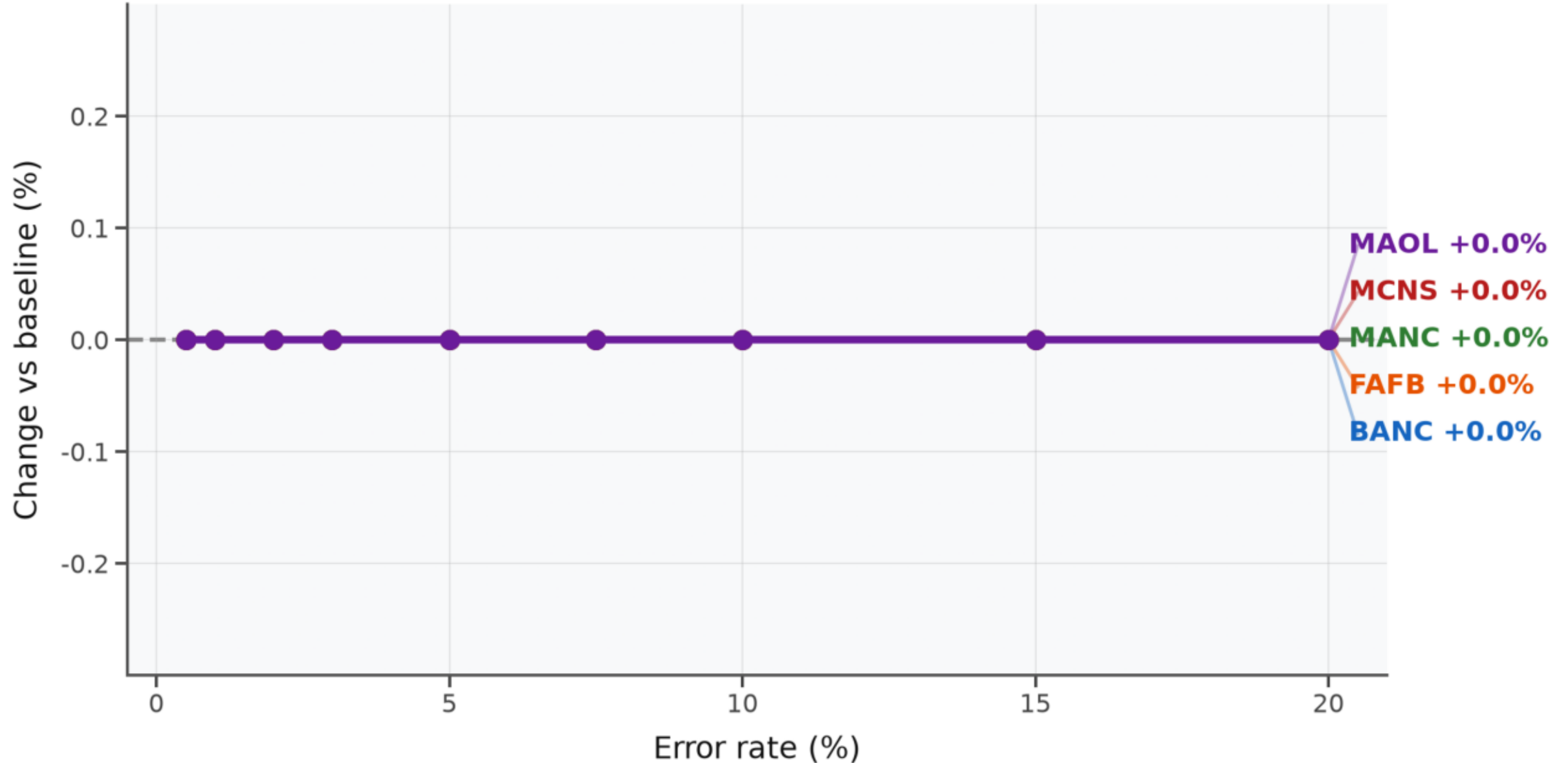


Figure 11 · Synapse count measurement — Edge count. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Synapse Count Noise — Total synapses

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

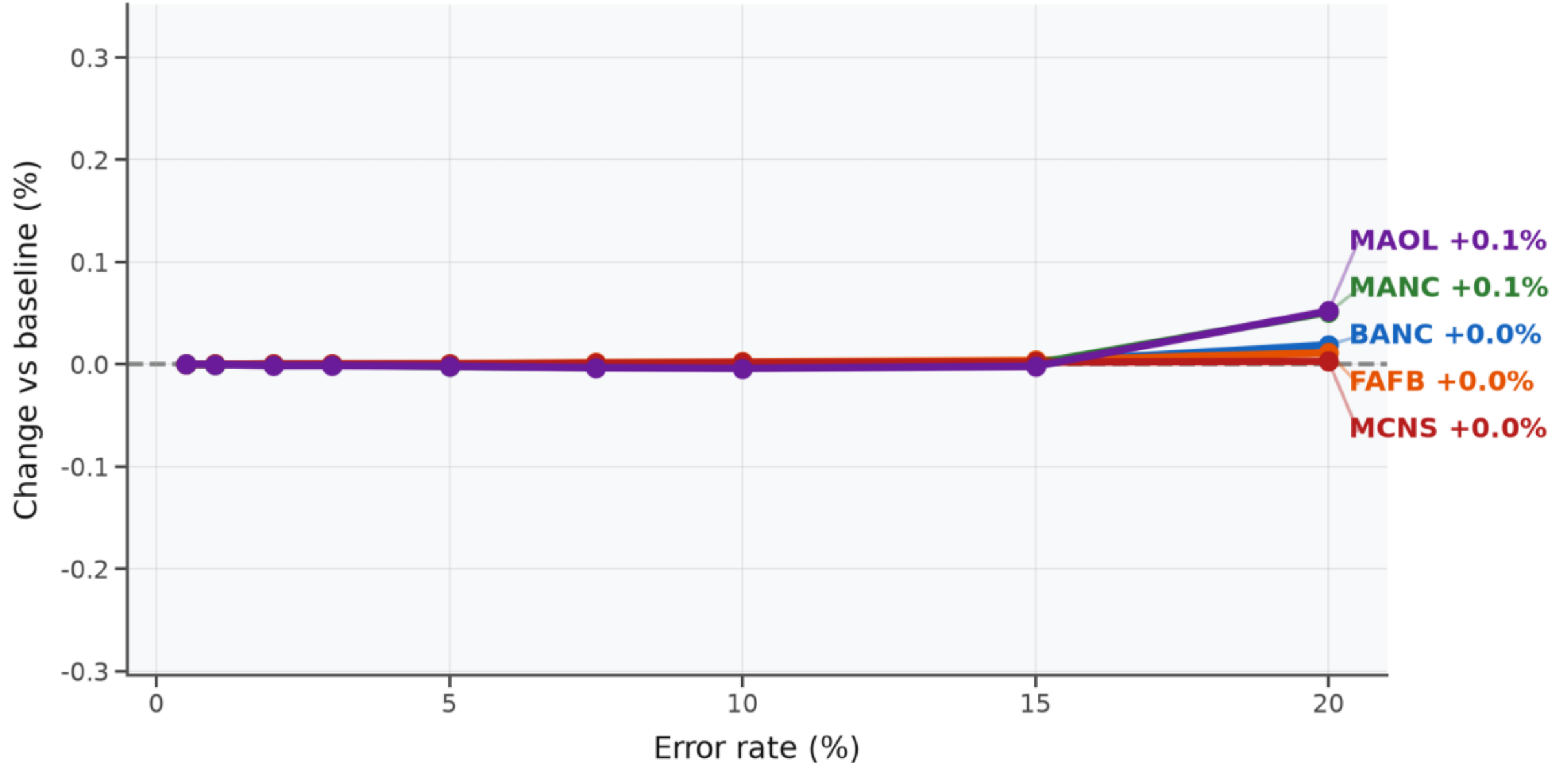


Figure 12 · Synapse count measurement — Total synapses. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Synapse Count Noise — Mean total degree

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

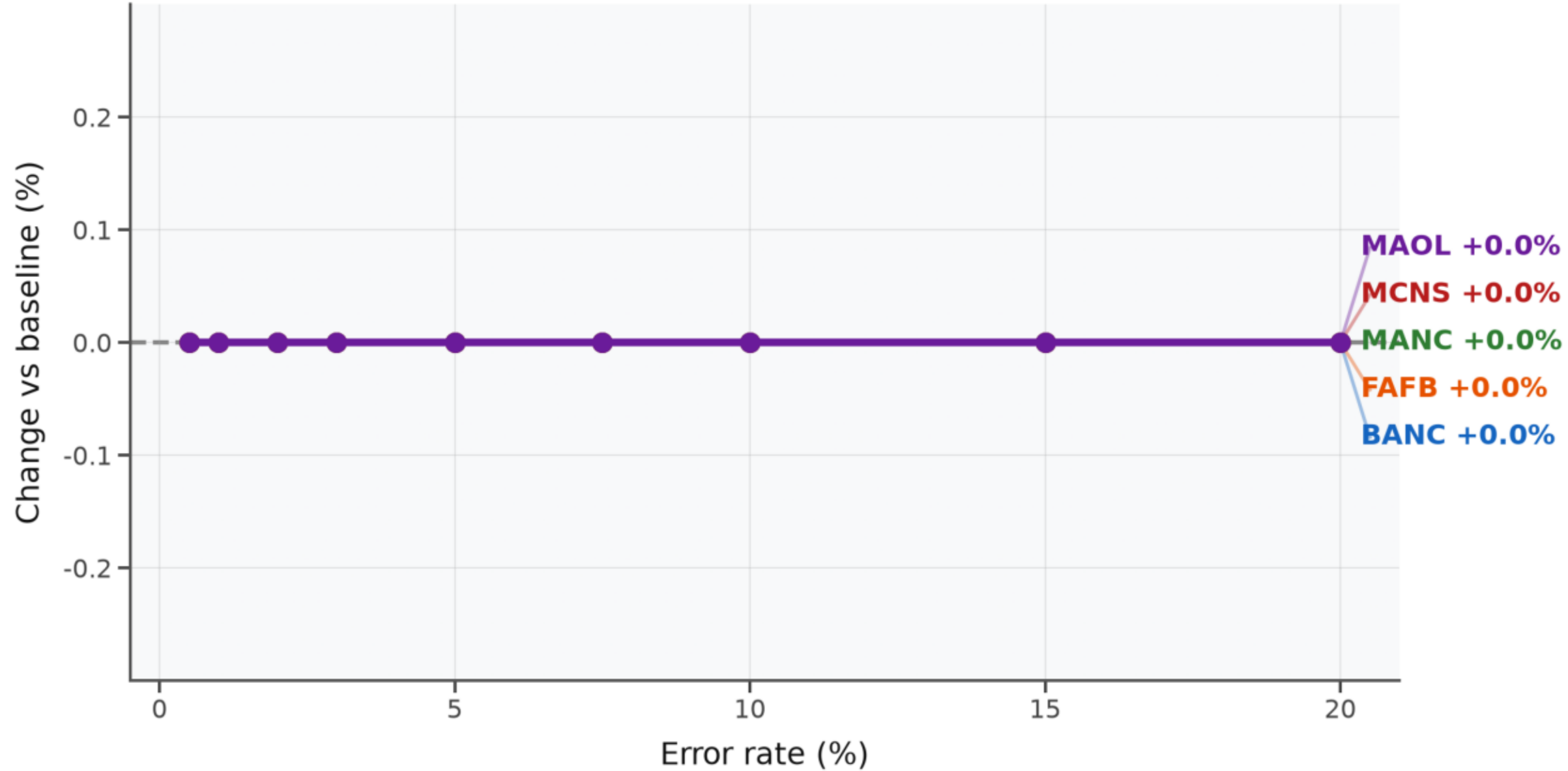


Figure 13 · Synapse count measurement — Mean total degree. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Synapse Count Noise — Largest weak component

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

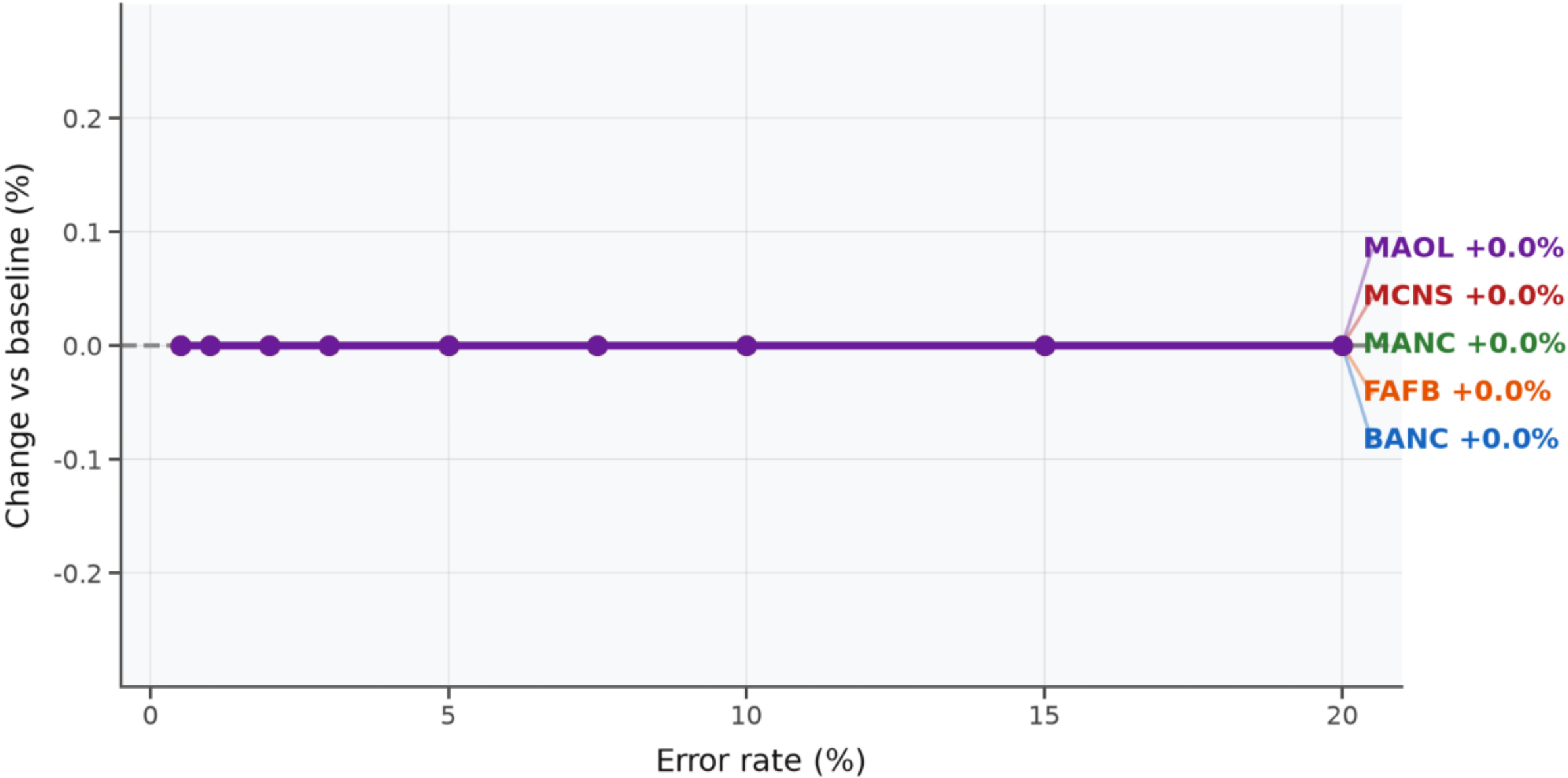


Figure 14 · Synapse count measurement — Largest weak component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Synapse Count Noise — Reciprocity

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

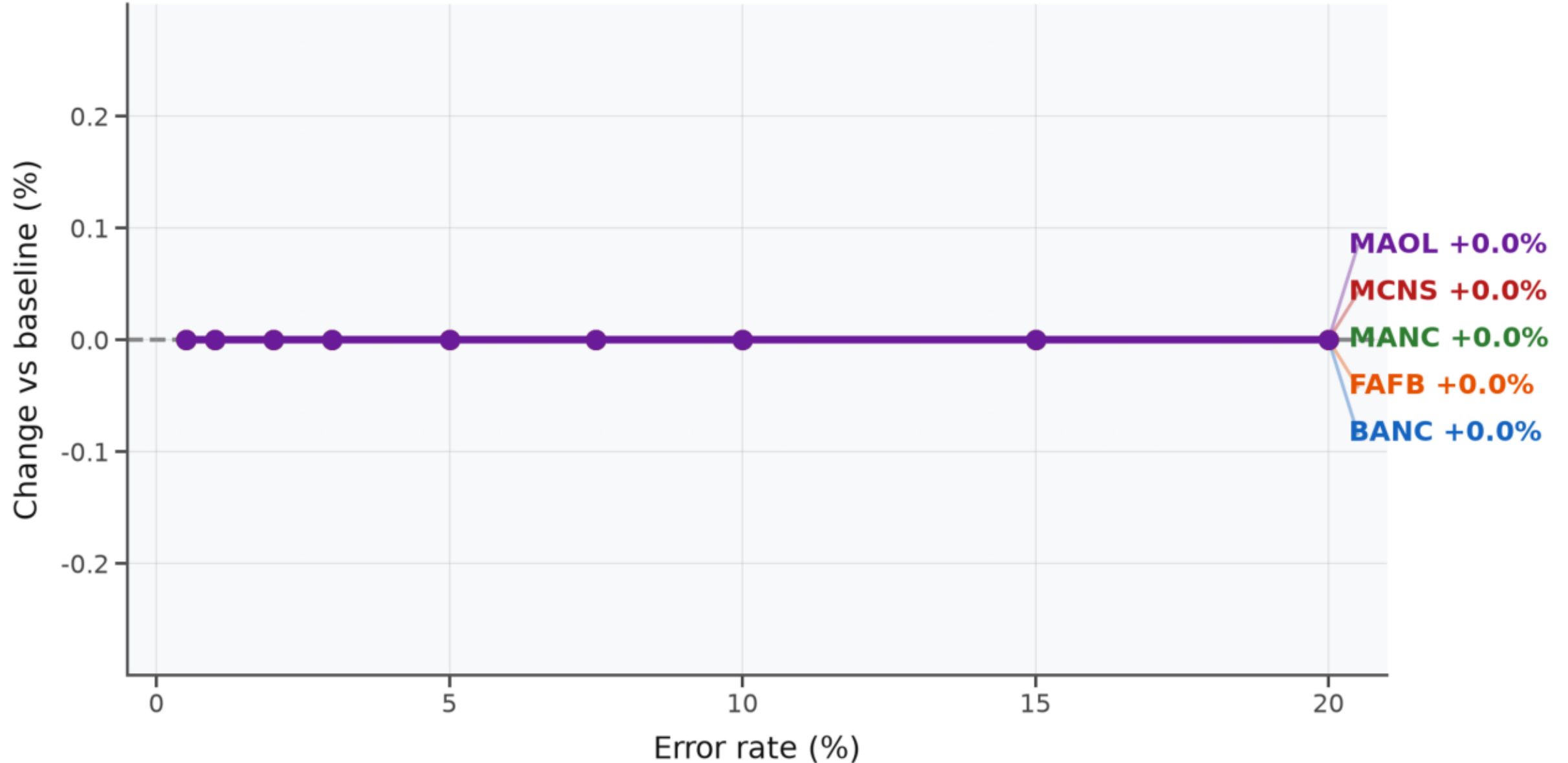


Figure 15 : Synapse count measurement — Reciprocity. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Split Errors — Edge count

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

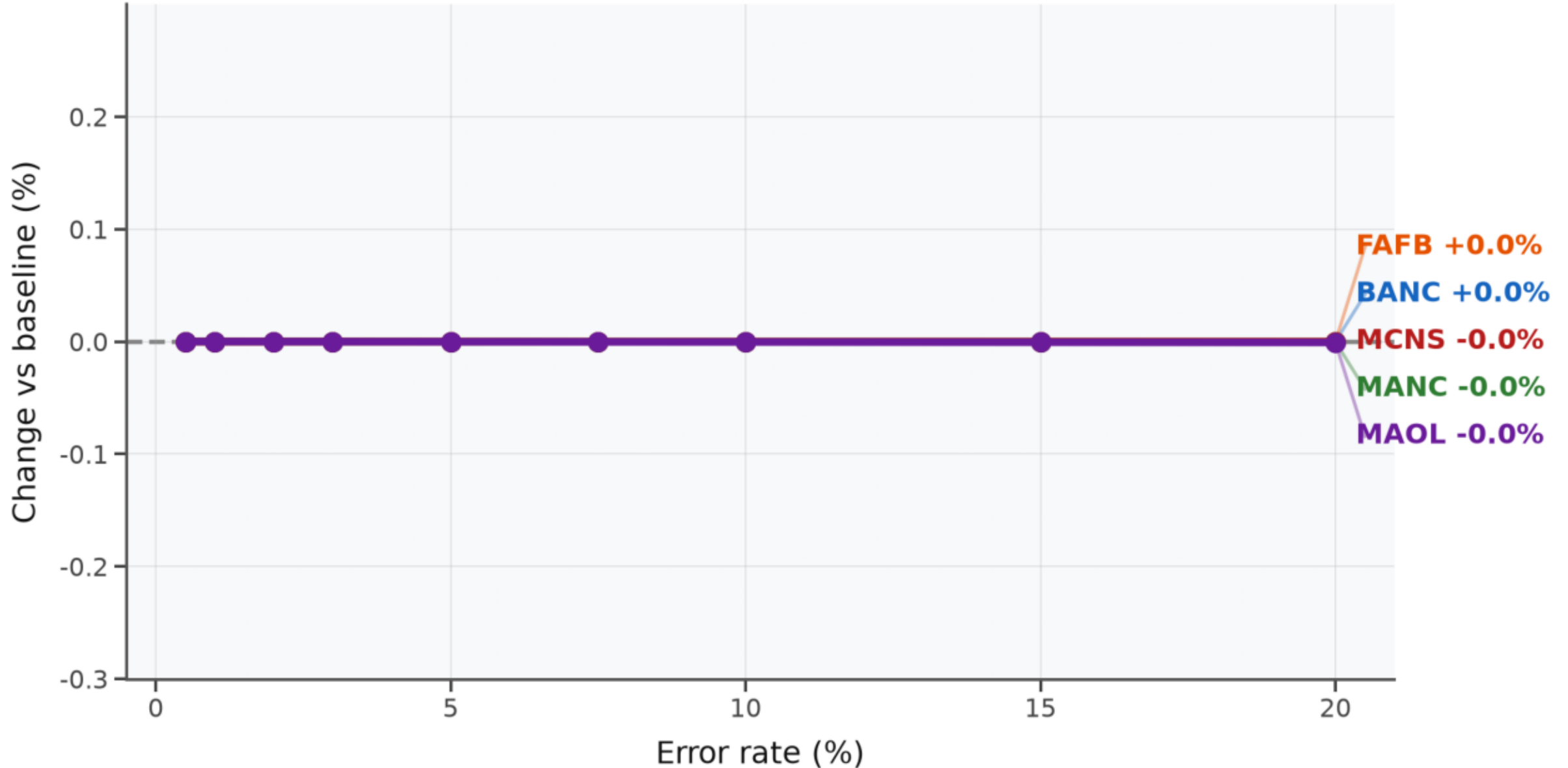


Figure 16 · Split errors — Edge count. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Split Errors — Total synapses

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

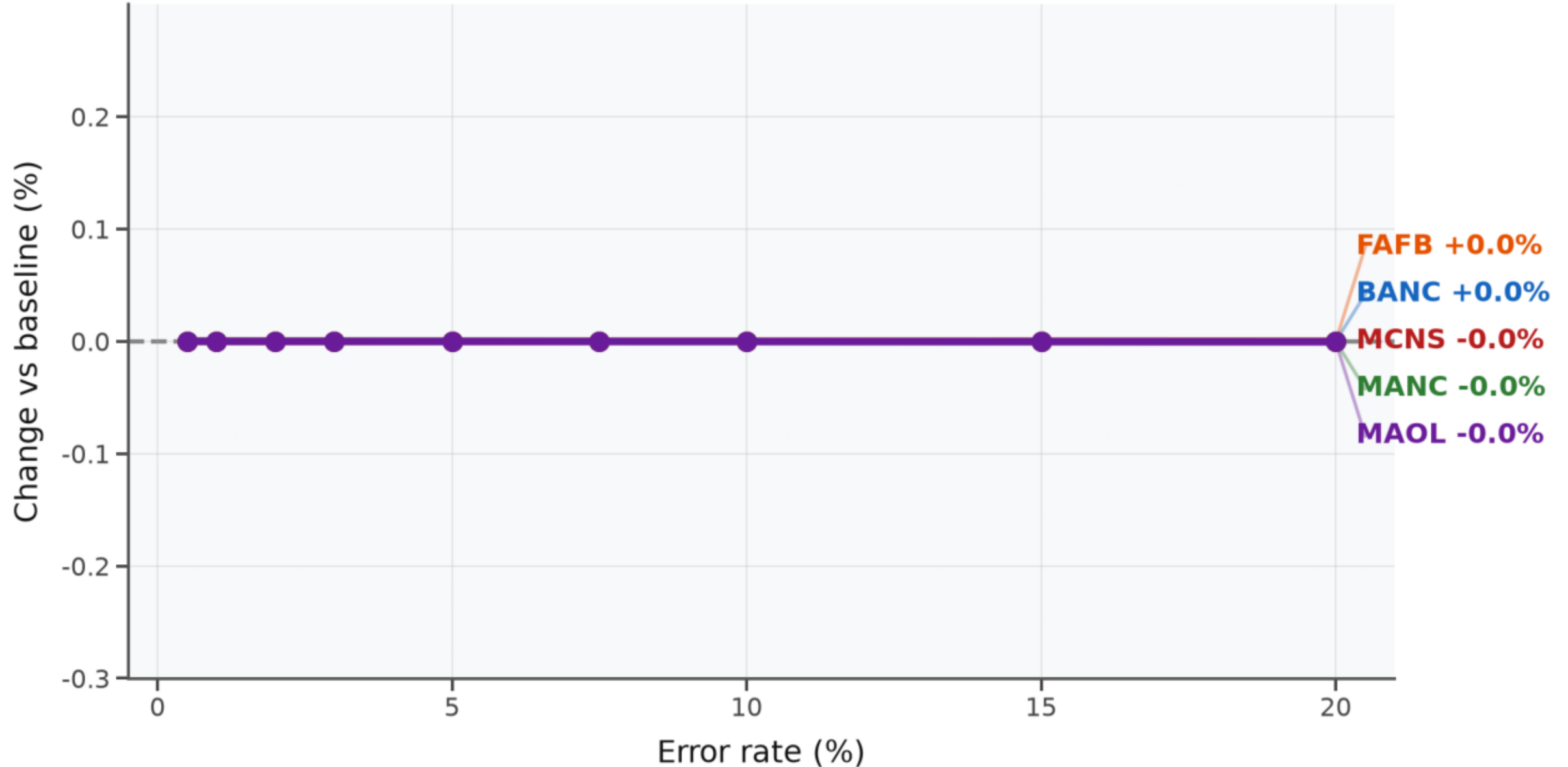


Figure 17 · Split errors — Total synapses. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Split Errors — Mean total degree

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

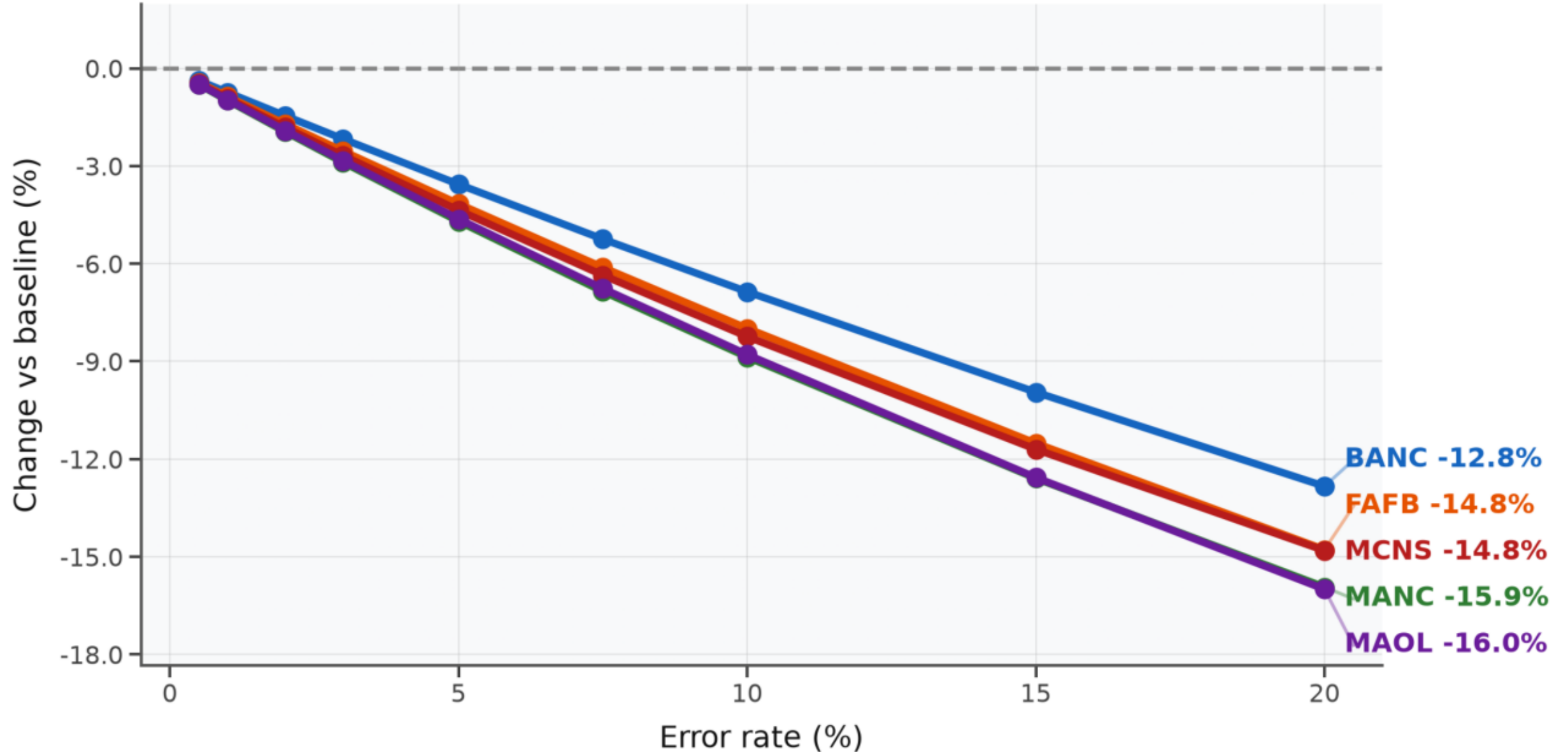


Figure 18 · Split errors — Mean total degree. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Split Errors — Largest weak component

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

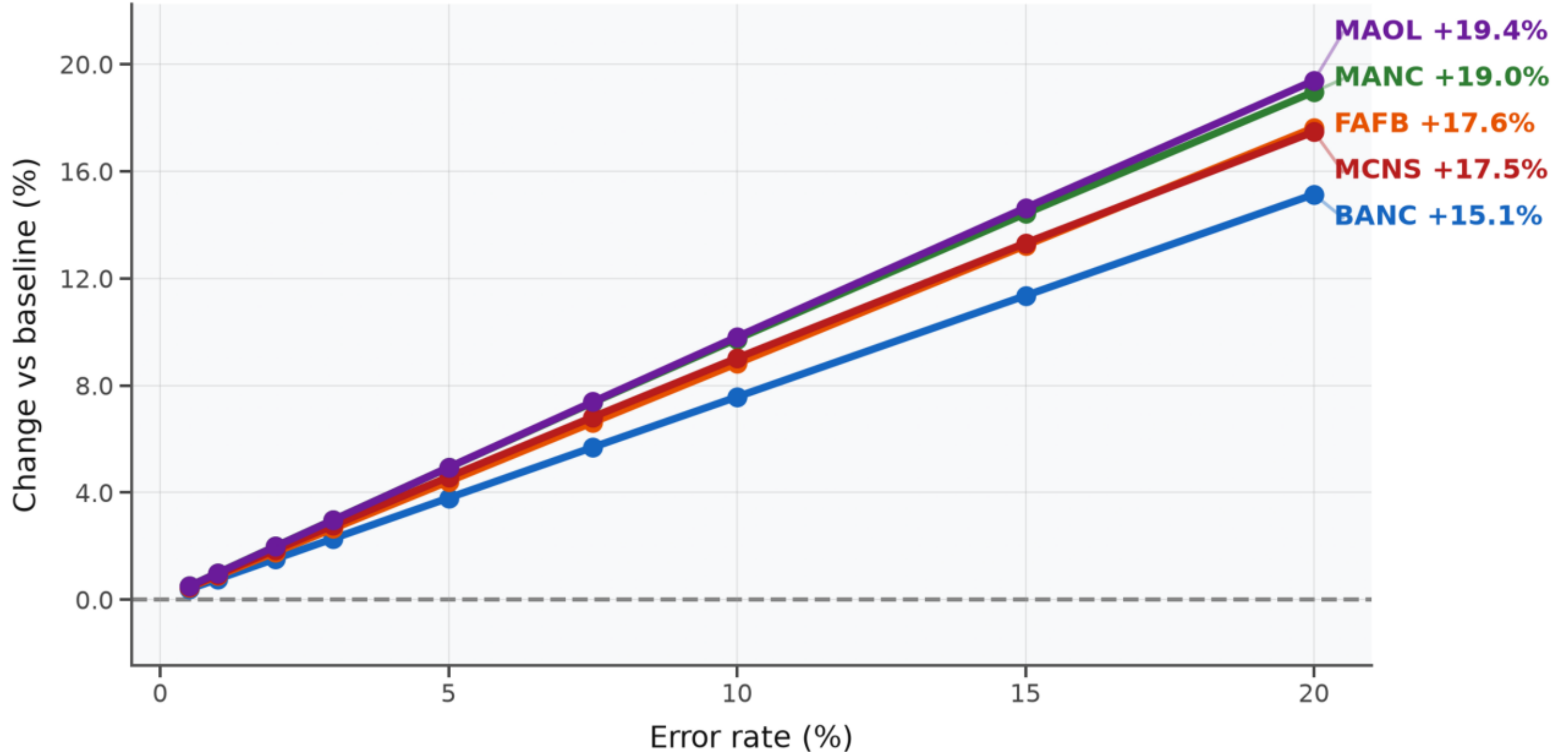


Figure 19 · Split errors — Largest weak component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Split Errors — Reciprocity

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

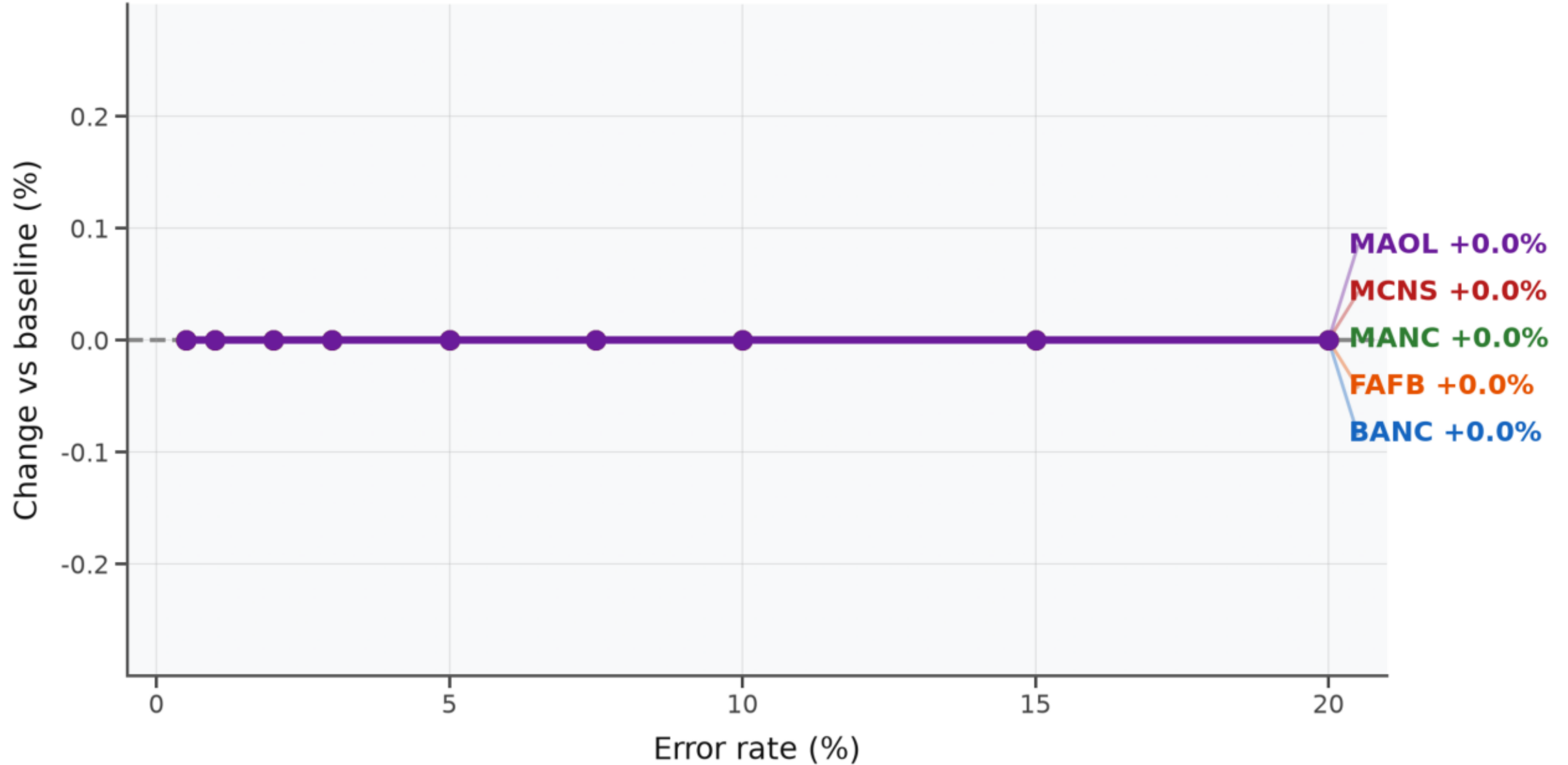


Figure 20 · Split errors — Reciprocity. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Merge Errors — Edge count

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

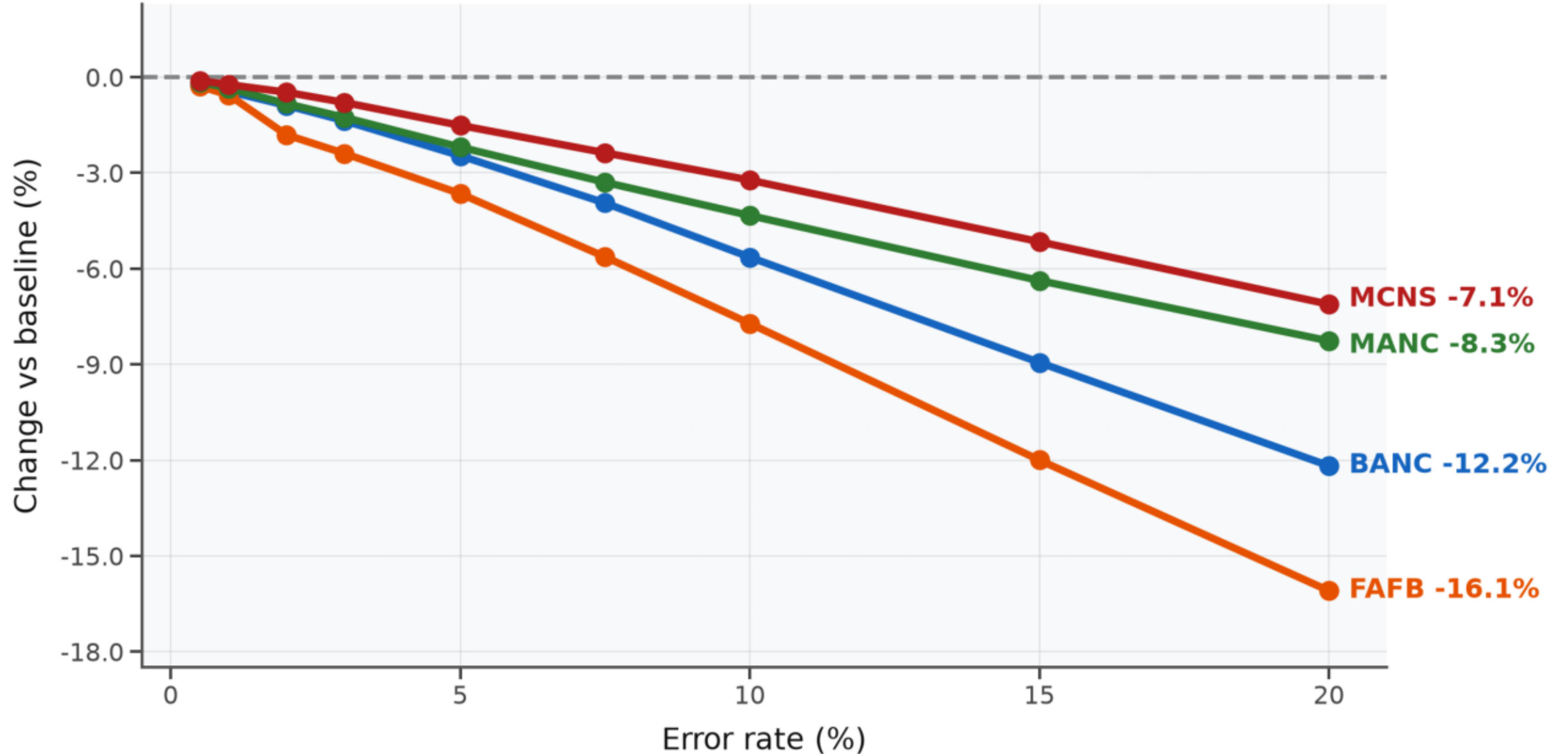


Figure 21 · Merge errors — Edge count. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Merge Errors — Total synapses

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

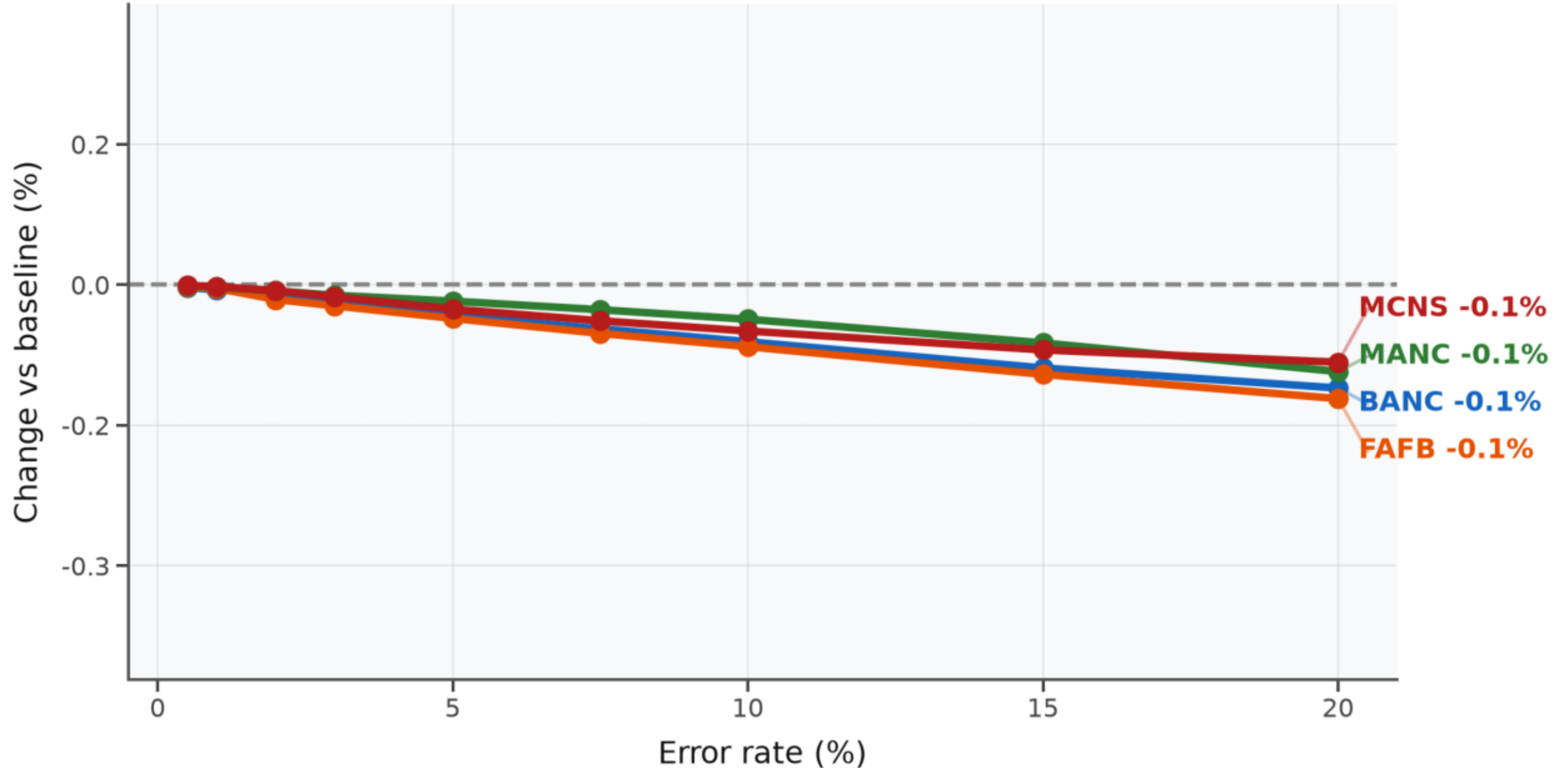


Figure 22 · Merge errors — Total synapses. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Merge Errors — Mean total degree

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

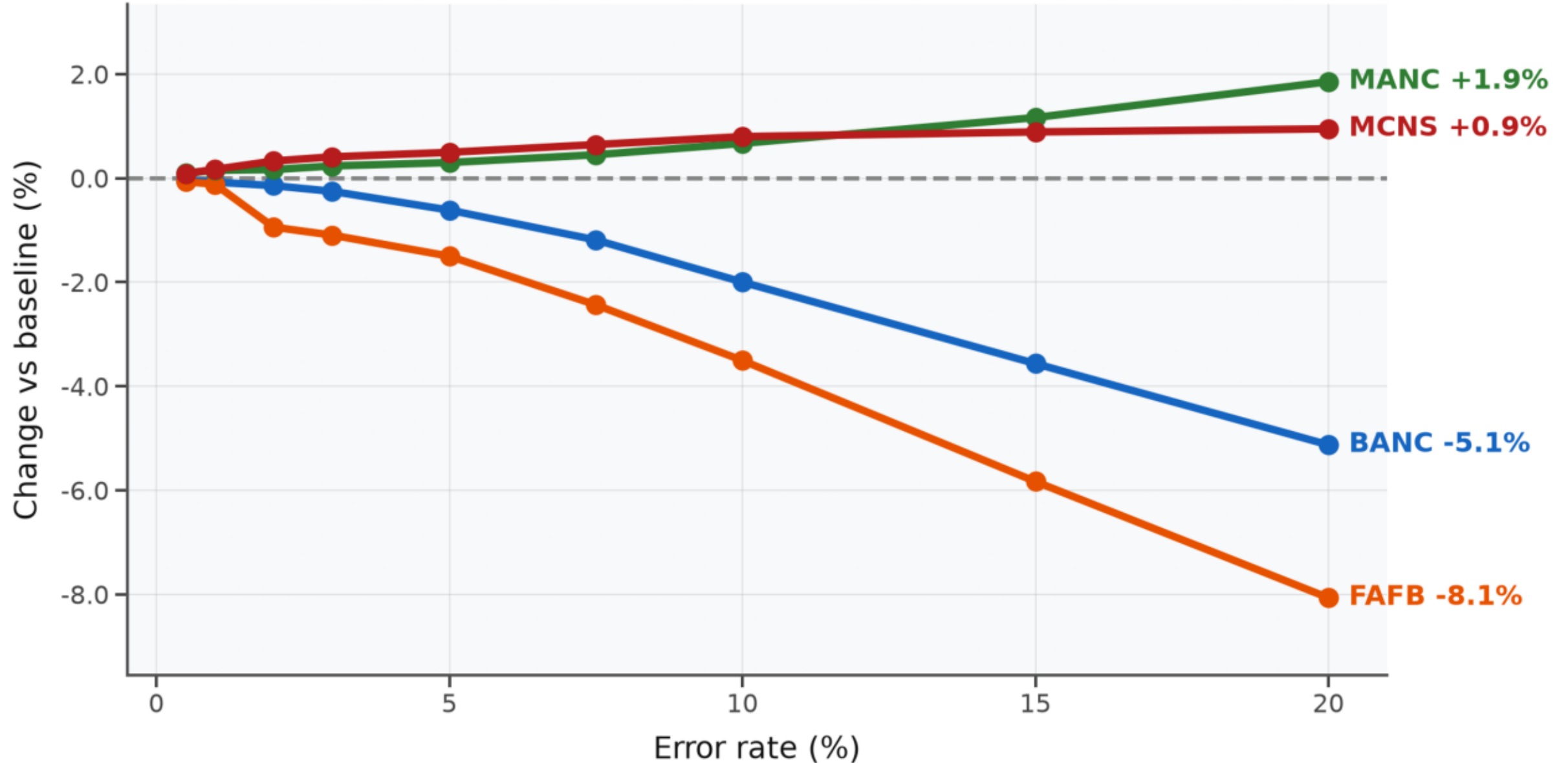


Figure 23 · Merge errors — Mean total degree. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Merge Errors — Largest weak component

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

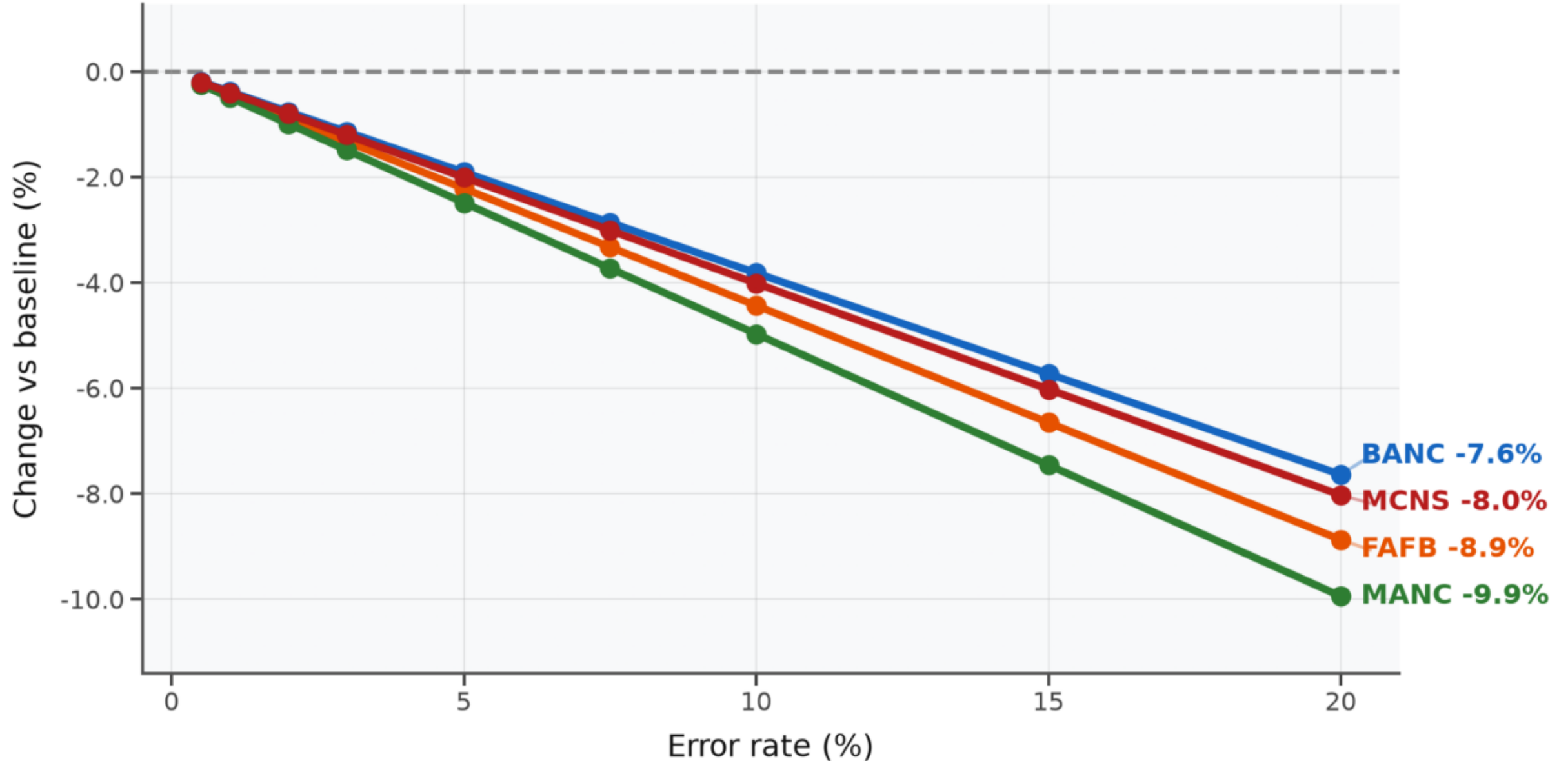


Figure 24 · Merge errors — Largest weak component. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

Merge Errors — Reciprocity

y = % change vs 0 % baseline (mean over replicate trials). Dashed line = no change.

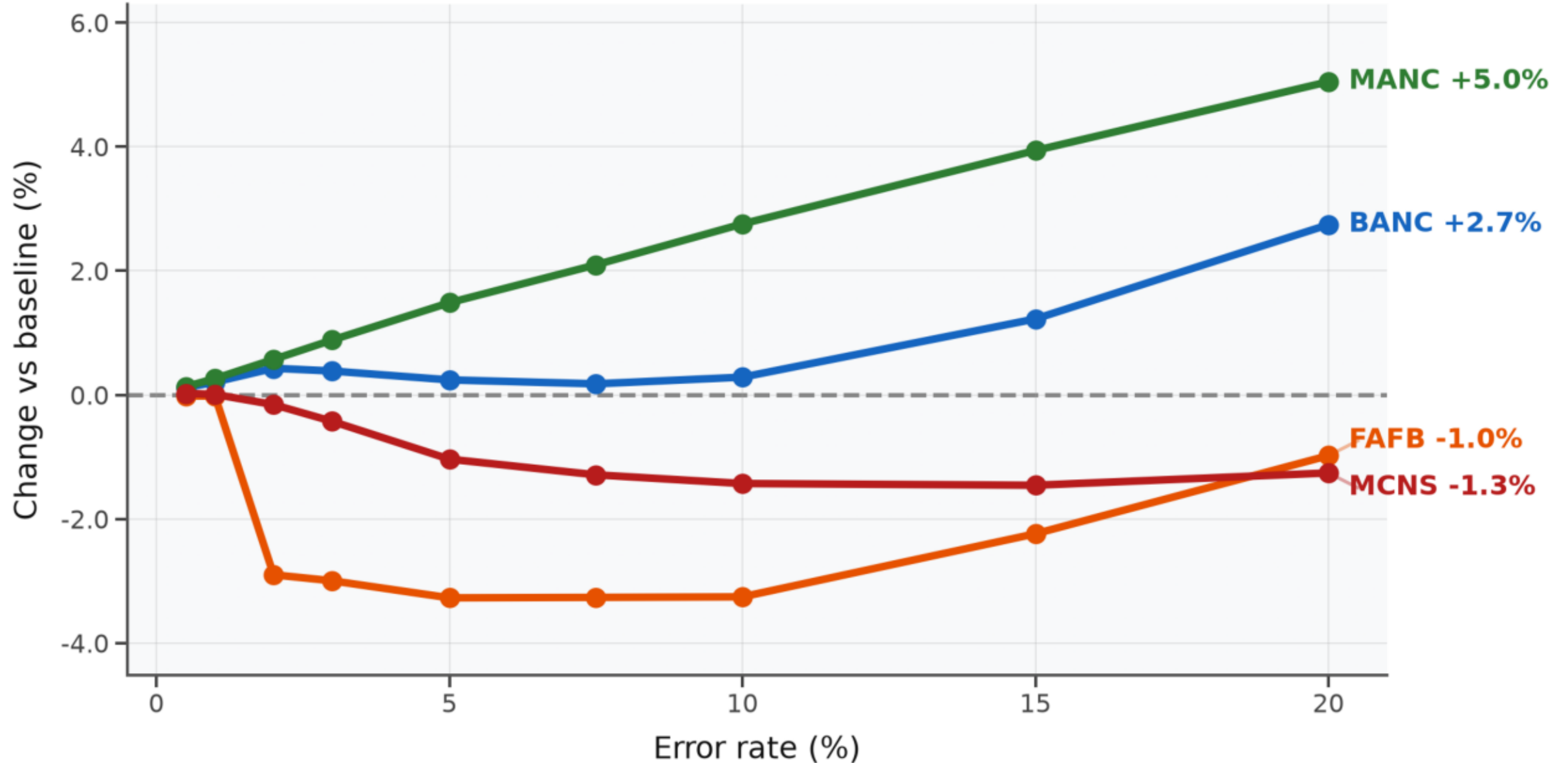


Figure 25 · Merge errors — Reciprocity. Each line represents a dataset's response to the error model. Direct endpoint labelling replaces the legend.

PageRank Preservation — Missed Synapses

Pearson correlation of perturbed vs baseline PageRank vectors (synapse-weighted, damping 0.85).

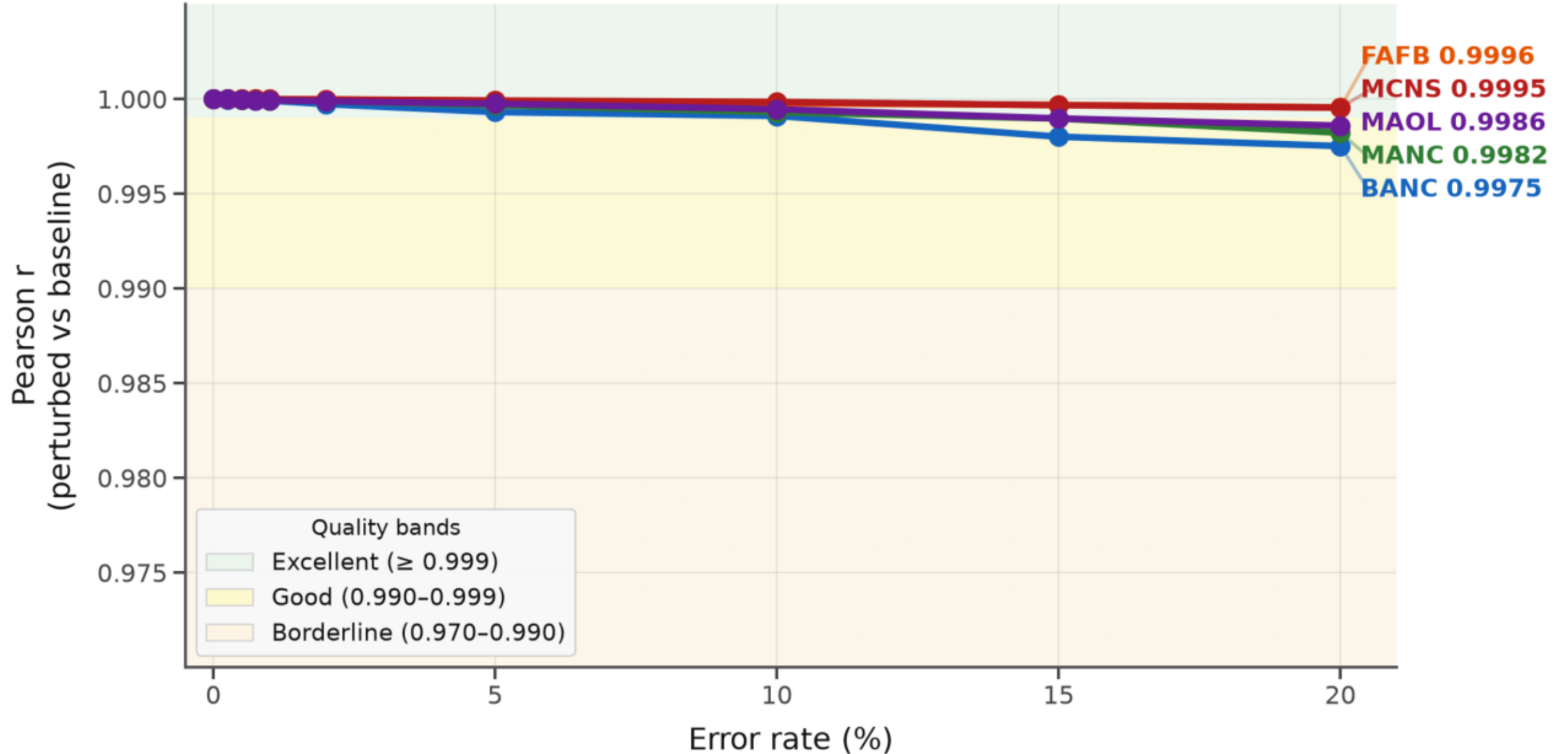


Figure 26 · PageRank preservation — Missed synapses. Coloured bands: green ≥ 0.999 (excellent), yellow $0.990-0.999$ (good), orange $0.970-0.990$ (borderline), red < 0.970 (degraded).

PageRank Preservation — False Synapses

Pearson correlation of perturbed vs baseline PageRank vectors (synapse-weighted, damping 0.85).

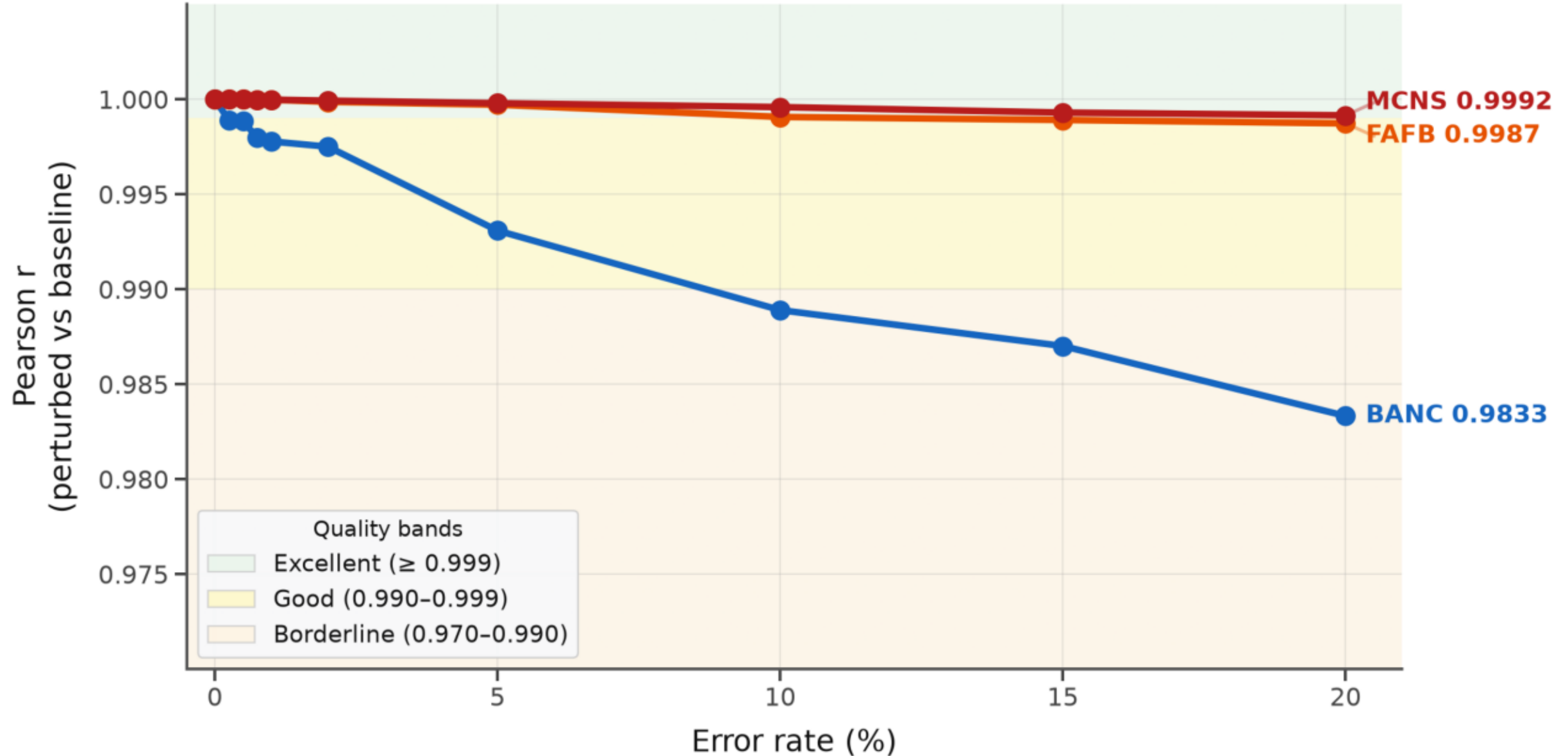


Figure 27 · PageRank preservation — False synapses. Coloured bands: green ≥ 0.999 (excellent), yellow $0.990-0.999$ (good), orange $0.970-0.990$ (borderline), red < 0.970 (degraded).

PageRank Preservation — Synapse Count Noise

Pearson correlation of perturbed vs baseline PageRank vectors (synapse-weighted, damping 0.85).

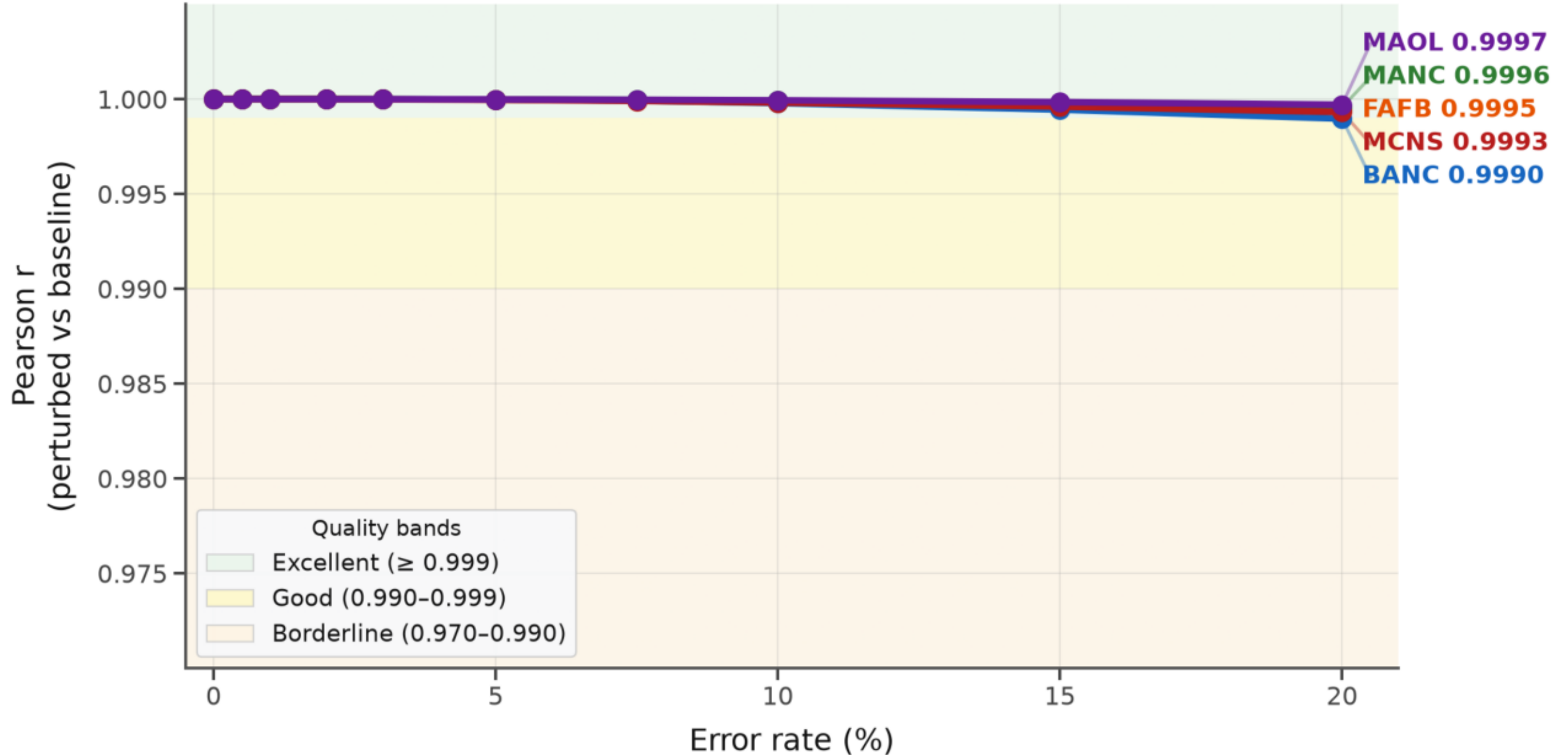


Figure 28 · PageRank preservation — Synapse count measurement. Coloured bands: green ≥ 0.999 (excellent), yellow $0.990-0.999$ (good), orange $0.970-0.990$ (borderline), red < 0.970 (degraded).

PageRank Preservation — Split Errors

Pearson correlation of perturbed vs baseline PageRank vectors (synapse-weighted, damping 0.85).

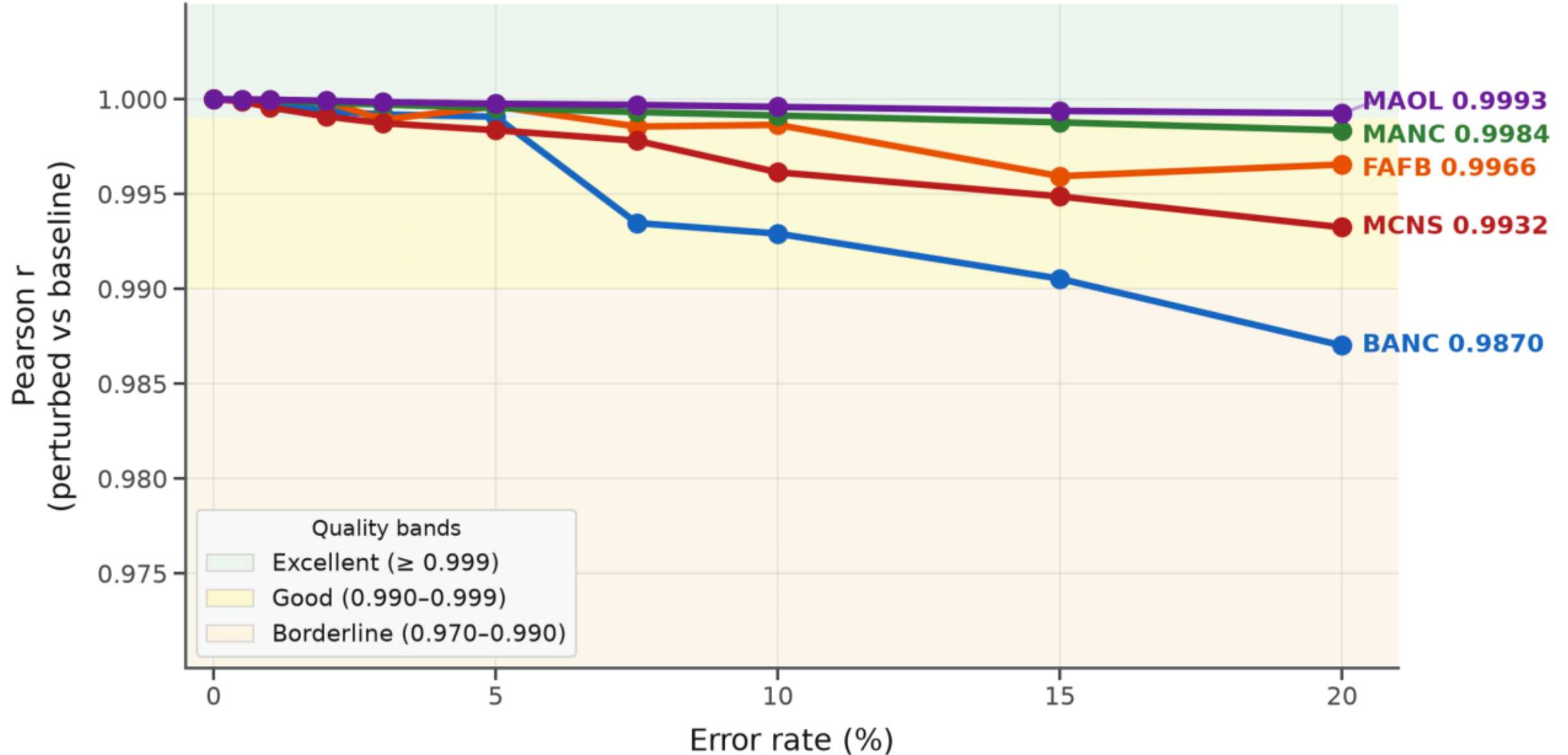


Figure 29 · PageRank preservation — Split errors. Coloured bands: green ≥ 0.999 (excellent), yellow $0.990-0.999$ (good), orange $0.970-0.990$ (borderline), red < 0.970 (degraded).

PageRank Preservation — Merge Errors

Pearson correlation of perturbed vs baseline PageRank vectors (synapse-weighted, damping 0.85).

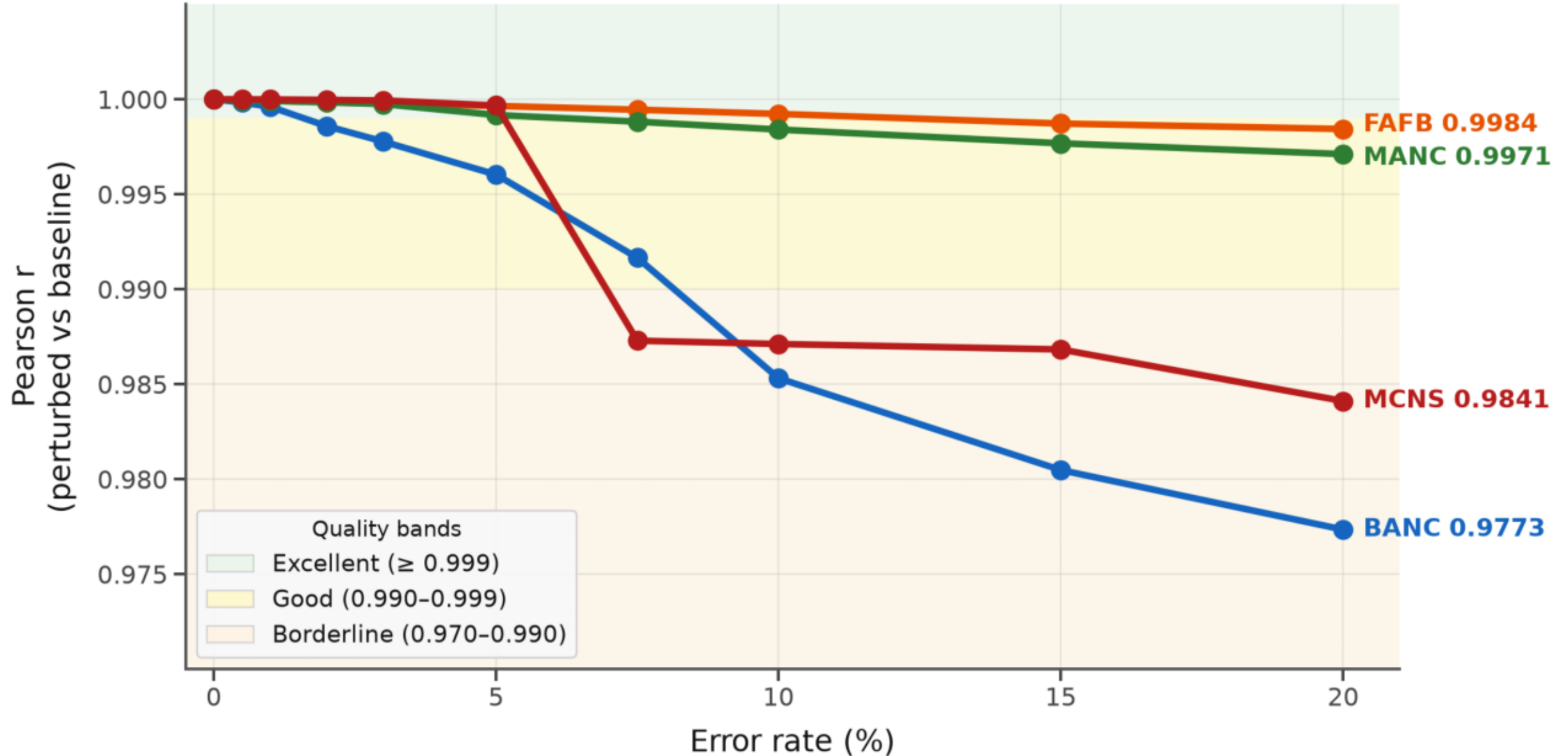


Figure 30 · PageRank preservation — Merge errors. Coloured bands: green ≥ 0.999 (excellent), yellow 0.990–0.999 (good), orange 0.970–0.990 (borderline), red < 0.970 (degraded).

§5 · Results — Worst-case Impact per Error Model

Worst-case impact of each error model across all FlyWire connectomes
(Maximum |% change| over all error rates; darker = more disruption; assortativity excluded — near-zero baseline)

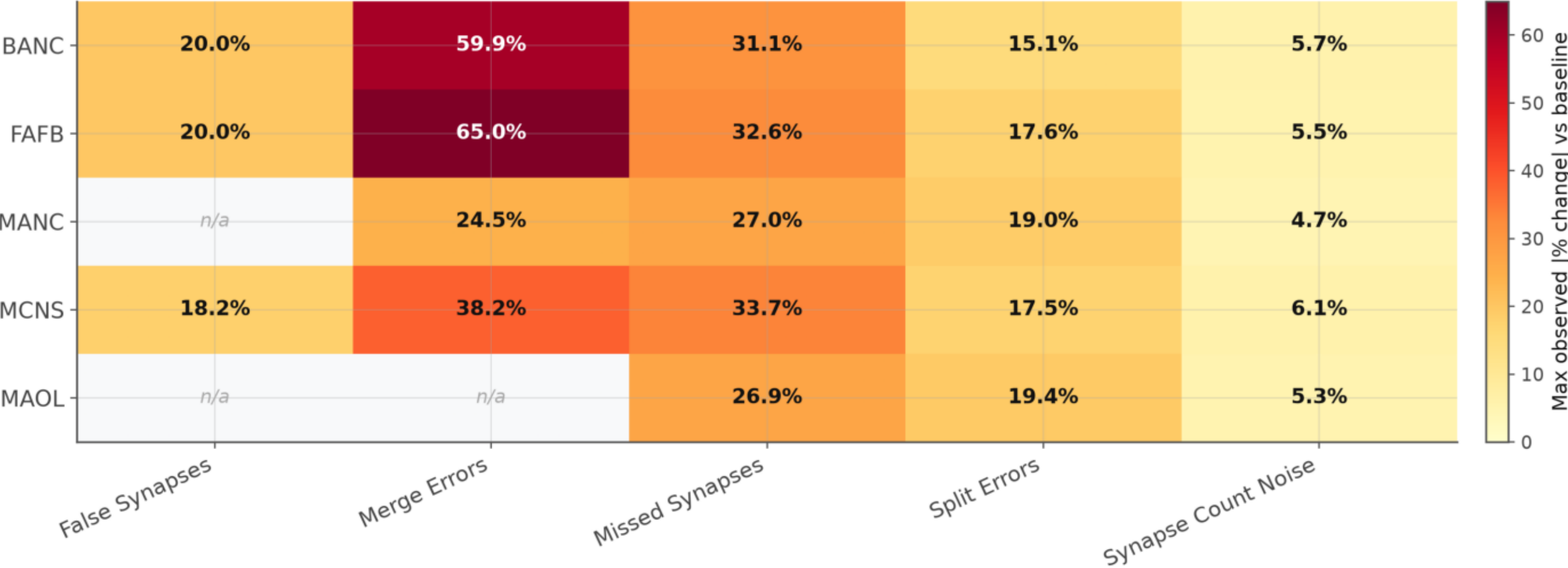


Figure 31 · Maximum observed |% change| per error model across all rates and datasets (assortativity excluded — near-zero baseline artefact). Darker cells indicate more disruption.

§5.5 · Effect Sizes at 20 % Error

Mean % change vs baseline across all datasets for each error model

Table 5 · % change at the maximum tested error rate (20 %)

Error model	Edge count	Total synapses	Weight variance	Mean degree	Largest WCC	Reciprocity	Assortativity*	PageRank r
Missed synapses	-4.9%	-20.0%	-30.3%	-4.9%	-0.0%	-2.9%	-3.4%	0.999
False synapses	+19.4%	+7.6%	-13.8%	+19.4%	+0.0%	+5.8%	+134.9%	0.994
Synapse count measurement	+0.0%	+0.0%	+5.5%	+0.0%	+0.0%	+0.0%	+0.0%	0.999
Split errors	-0.0%	-0.0%	+0.0%	-14.9%	+17.7%	+0.0%	+0.3%	0.995
Merge errors	-10.9%	-0.1%	+46.9%	-2.6%	-8.6%	+1.4%	-2.2%	0.989

*Assortativity has a near-zero baseline (−0.03 to −0.06), so its % change is inflated and should be read as an absolute movement, not a percentage. PageRank r is the mean Pearson correlation across datasets (baseline = 1.0 by construction); Spearman and top-100 overlap are reported in §5.5b.

Spread across datasets (min ... max)

Missed synapses: Edge count -9.7...-0.0%; Total synapses -20.0...-20.0%; Weight variance -33.7...-26.9%; Mean total degree -9.7...-0.0%; Largest weak comp. -0.1...+0.0%; Reciprocity -6.9...-0.0%; Assortativity* -9.3...+0.5%

False synapses: Edge count +18.2...+20.0%; Total synapses +6.3...+10.0%; Weight variance -15.0...-12.6%; Mean total degree +18.2...+20.0%; Largest weak comp. +0.0...+0.0%; Reciprocity +2.2...+8.0%; Assortativity* +16.1...+356.6%

Synapse count measurement: Edge count +0.0...+0.0%; Total synapses +0.0...+0.1%; Weight variance +4.7...+6.1%; Mean total degree +0.0...+0.0%; Largest weak comp. +0.0...+0.0%; Reciprocity +0.0...+0.0%; Assortativity* +0.0...+0.0%

Split errors: Edge count -0.0...+0.0%; Total synapses -0.0...+0.0%; Weight variance +0.0...+0.0%; Mean total degree -16.0...-12.8%; Largest weak comp. +15.1...+19.4%; Reciprocity +0.0...+0.0%; Assortativity* -10.0...+5.0%

Merge errors: Edge count -16.1...-7.1%; Total synapses -0.1...-0.1%; Weight variance +24.5...+65.0%; Mean total degree -8.1...+1.9%; Largest weak comp. -9.9...-7.6%; Reciprocity -1.3...+5.0%; Assortativity* -10.1...+3.2%

Missed synapses remove exactly 20 % of synapses at the 20 % rate, cutting weight variance by 30 % but only deleting the ~5 % of edges whose weight reached zero — and PageRank vectors stay highly correlated (Pearson 0.999, Spearman 0.998, top-100 overlap 0.98 for missed synapses at 20 %), so most high-centrality neurons keep their rank. False synapses add ~19 % more edges, but the new low-weight edges dilute weight variance by ~14 %. Synapse-count noise moves only the weight variance (+5.5 %). Split errors preserve edges but spread them over more neurons (mean degree −15 %, largest weak component +18 %). Merge errors are the most invasive: weight variance +47 %, edges −11 %, largest component −9 %.

§5.5b · PageRank Preservation at 20 % Error

Pearson, Spearman and top-100 overlap — all three measures present in the exports

Table 5b · PageRank preservation at 20 % error — mean across datasets (min-max across datasets)

Error model	Pearson r	Spearman ρ	Top-100 overlap
Missed synapses	0.999 (0.998–1.000)	0.998 (0.996–0.999)	0.980 (0.970–0.986)
False synapses	0.994 (0.983–0.999)	0.971 (0.962–0.980)	0.951 (0.892–0.982)
Synapse count measurement	0.999 (0.999–1.000)	0.999 (0.998–1.000)	0.976 (0.970–0.986)
Split errors	0.995 (0.987–0.999)	0.976 (0.969–0.986)	0.956 (0.942–0.978)
Merge errors	0.989 (0.977–0.998)	0.989 (0.986–0.991)	0.924 (0.860–0.960)

Pearson correlation measures vector similarity, not rank preservation. Spearman correlation measures rank-order agreement, and top-100 overlap measures how many of the 100 highest-PageRank neurons in the baseline remain in the top 100 after perturbation. Across all models Pearson $r \geq 0.977$ and Spearman ≥ 0.96 at 20 % error, so PageRank vectors remain highly correlated and rank order is largely, but not entirely, retained. The largest displacements occur on BANC: under merge errors the top-100 overlap falls to 0.86 and under false synapses to 0.89, i.e. roughly one in seven of the top hubs is displaced from the baseline top 100.

§5.6 · Per-Dataset Breakdown at Peak Error Rate

How variation across connectomes reinforces the headline conclusions

Table 6 · Edge count and total-synapse change per dataset at 20 % error, with PageRank Pearson r (FAFB and MANC merge rows average 2 and 1 trials per rate; all other rows average 5)

Error model	Dataset	Edge count Δ%	Total synapses Δ%	PageRank r
Missed synapses	BANC	-3.2%	-20.0%	0.998
Missed synapses	FAFB	-2.5%	-20.0%	1.000
Missed synapses	MANC	-9.7%	-20.0%	0.998
Missed synapses	MCNS	-0.0%	-20.0%	1.000
Missed synapses	MAOL	-8.9%	-20.0%	0.999
False synapses	BANC	+20.0%	+10.0%	0.983
False synapses	FAFB	+20.0%	+6.6%	0.999
False synapses	MCNS	+18.2%	+6.3%	0.999
Synapse count measurement	BANC	+0.0%	+0.0%	0.999
Synapse count measurement	FAFB	+0.0%	+0.0%	0.999
Synapse count measurement	MANC	+0.0%	+0.1%	1.000
Synapse count measurement	MCNS	+0.0%	+0.0%	0.999
Synapse count measurement	MAOL	+0.0%	+0.1%	1.000
Split errors	BANC	+0.0%	+0.0%	0.987
Split errors	FAFB	+0.0%	+0.0%	0.997
Split errors	MANC	-0.0%	-0.0%	0.998
Split errors	MCNS	-0.0%	-0.0%	0.993
Split errors	MAOL	-0.0%	-0.0%	0.999
Merge errors	BANC	-12.2%	-0.1%	0.977
Merge errors	FAFB	-16.1%	-0.1%	0.998
Merge errors	MANC	-8.3%	-0.1%	0.997
Merge errors	MCNS	-7.1%	-0.1%	0.984

This table shows that the cross-dataset means in Table 5 do not hide large disagreements between connectomes. The −4.9 % average edge loss under missed synapses spans −0.0 % (MCNS) to −9.7 % (MANC): datasets whose edges carry fewer synapses lose edges faster, because an edge survives only if at least one of its synapses survives (Table 6b). Merge errors reduce edge count in every dataset at 20 % (FAFB −16.1 % to MCNS −7.1 %); no dataset shows an increase.

§5.6b · Why Edge Loss Differs Across Connectomes

Edge-weight distribution vs edge loss at 20 % synapse loss (missed-synapse model)

Table 6b · Baseline edge-weight distribution vs edge loss under the missed-synapse model at 20 %

Dataset	Mean synapses/edge	Median synapses/edge	Edge count $\Delta\%$ at 20 % synapse loss
BANC	5.9	4	-3.2%
FAFB	9.5	6	-2.5%
MANC	5.0	2	-9.7%
MCNS	14.4	9	-0.0%
MAOL	3.9	2	-8.9%

An edge survives a missed-synapse run only if at least one of its synapses survives, so at a fixed synapse-loss rate the fraction of edges deleted is largest for datasets whose edges carry the fewest synapses. The median synapses/edge tracks the observed edge loss monotonically across the five connectomes: MANC and MAOL (median 2) lose -9.7% and -8.9% of edges, BANC (median 4) -3.2% , FAFB (median 6) -2.5% , and MCNS (median 9) -0.0% . This is a descriptive relationship across five connectomes ($n = 5$); it is consistent with the binomial-deletion mechanism but is not claimed as a general law.

§6 · Conclusions

Practical implications for segmentation benchmarking

- 1. Disproportionate Impact of Connection Loss: Missed synapses remove synapses proportionally with the error rate and erode weight variance by 30 % at 20 %, while false synapses primarily inflate edge counts. Algorithmic benchmarking should prioritize missed-synapse detection.
- 2. Resilience to Measurement Noise: Synapse-count uncertainty perturbs weight statistics but leaves the overarching topology intact. Graph analyses unweighted by synapse count remain unaffected.
- 3. Structural Shifts via Mis-segmentation: Split and merge errors alter the neuron count rather than just edge topology. Analyses treating neurons as fundamental units (e.g., community detection, PageRank) are highly sensitive to these errors.
- 4. PageRank Robustness: Despite local topological degradation, PageRank vectors remain highly correlated at a 20 % error rate (Pearson $r \geq 0.977$ and Spearman ≥ 0.96 for every model and dataset). Most, but not all, high-centrality neurons retain their rank: top-100 hub overlap stays ≥ 0.86 , with the largest displacements under false and merge errors on BANC (Table 5b). Local measures (degree, component size) act as earlier indicators of structural decay.
- 5. Consistency Across the Evaluated Connectomes: The five evaluated *Drosophila* connectomes exhibit qualitatively similar responses to identical error models. Broader generalization is not claimed: all datasets come from one species, and coverage is partial (Table 2).

Caveats

FAFB and MANC merge-error results rest on 2 and 1 trials per rate (partial runs in the source archives); MANC and MAOL have no false-synapse run. All other cells use 5 replicate trials, and trial-to-trial SD of the reported % changes is below 0.15 pp for every 5-trial cell. The %-change framing inflates metrics with near-zero baselines (assortativity) — always check the raw direction and magnitude. The five error models are computational simulations: their error distributions are not empirically validated against real reconstruction-error ground truth (no such data is part of this repository), so the results describe how the modelled error classes perturb graph structure, not measured segmentation errors. 1,030 trials in total; every number in this report was recomputed from the per-trial CSVs.

§7 · Reproducibility

How to regenerate this report from source data

The whole report — CSVs, figures and this PDF — is produced by three scripts in the analysis/ folder. Run them in order from the repository root:

```
$ python analysis/01_load_and_verify.py

$ python analysis/01_load_and_verify.py
loaded 6180 analysis-rows, 5 datasets, 5 error models, 10 rates, max 5 trials per cell

[1] COMPLETENESS
  trials per cell: min=1, max=5, cells=220
  PARTIAL FAFB/merge_errors: only 2 trials/rate
  PARTIAL MANC/merge_errors: only 1 trial/rate
  analyses per trial: min=6, max=6 (expect 6)
  non-SUCCESS rows: 0

[2] BASELINE INVARIANCE (0% error)
  BANC: baseline max rel diff across error models = 0.00e+00
  FAFB: baseline max rel diff across error models = 0.00e+00
  MANC: baseline max rel diff across error models = 0.00e+00
  MCNS: baseline max rel diff across error models = 0.00e+00
  MAOL: baseline max rel diff across error models = 0.00e+00

[3] INTERNAL CONSISTENCY
  in-degree mean vs edges/nodes: max rel err = 2.206e-16
  density vs edges/(n*(n-1)):    max rel err = 6.978e-13
```

Appendix · Statistical Notes

Known artefacts and how they are handled in this report

- Near-zero-baseline % change. Relative change is undefined in spirit when the baseline is near zero. Assortativity (baseline $-0.03\dots-0.06$) shows % swings of +135 % for a small absolute movement; we report the raw sign and magnitude and exclude the metric from the heatmap.
- Reciprocity baselines (0.13–0.29) are safe for % change; the biggest move is merge errors (+1.4 % mean).
- Mean-over-datasets. Table 5 averages % changes across the datasets that were run. Table 6 gives every dataset individually, and the spread (min...max) is printed under Table 5.
- Partial runs. FAFB merge (2 trials/rate), MANC merge (1 trial/rate); MANC/MAOL false synapses and MAOL merge not run. Figures simply omit missing lines.
- Deterministic vs stochastic. False-synapse edge additions are deterministic ($k = \text{round}(r \cdot E)$ every trial), so their trial-to-trial spread is zero — an identity, not a bug (verified in §4).